

ISC Class XII Mathematics

Chapter 4 Determinants

41 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

I. Basics of Determinants

1. Evaluating 2×2 determinants (numeric/trig)
2. Evaluating 3×3 determinants by expansion
3. Finding cofactors of elements
4. Solving for x from a determinant equation
5. Proving an identity by full expansion
6. Showing a determinant is independent of a variable
7. Cofactor expansion along a specified row

II. Properties of Determinants

8. $\text{Det.} = 0$ via identical/proportional rows-columns
9. Using $|kA| = k^n |A|$ and related properties
10. Solving for x using the row-sum trick
11. Proving one root via a row operation
12. Equality of determinants via row/column interchange
13. Common-factor row, then expanding reduced minor
14. Using an A.P. condition with row operations
15. Vandermonde-style factorisation identities
16. Symmetric/cyclic-function identities in a, b, c
17. Deriving an algebraic relation from $\text{det.} = 0$
18. Geometric proof (isosceles triangle) from $\text{det.} = 0$

III. Area of a Triangle

19. Area of a triangle from three vertices
20. Checking collinearity of three points
21. Unknown coordinate from a collinearity condition
22. Equation of a line through two points
23. Unknown parameter from a given triangle area

IV. Adjoint and Inverse

24. Finding the adjoint of a matrix
25. Singularity – finding an unknown for singular A
26. Finding A^{-1} using the adjoint formula
27. Determinant–adjoint relations
28. Verifying $A(\text{adj } A) = |A|I = (\text{adj } A)A$
29. Finding A^{-1} via a matrix polynomial identity
30. Solving a matrix equation using inverses
31. Proving A^{-1} equals a special form of A

V. Systems of Linear Equations

32. Solving a 2×2 system by matrix method
33. Consistency/inconsistency of a system
34. Solving a 3×3 system by matrix method
35. Reciprocal-substitution systems
36. Word problems solved as linear systems (7 examples)
37. Homogeneous systems
38. Condition for a unique solution

VI. Higher-Order / Competency-Focused

39. Higher-degree identity (a^2, b^2, c^2 entries)
40. Squared-determinant identity (equilateral triangle)
41. Proving $A^{-1} = A'$ via row dot-products

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2	Evaluating 3×3 determinants by expansion	–	Ex 4.1 Q9(ii)
3	Finding cofactors of elements	–	Ex 4.1 Q4(i)
4	Solving for x from a determinant equation	Yes	Ex 4.1 Q5(i); Q7(i) (alt.)
5	Proving an identity by full expansion	–	Ex 4.1 Q10
6	Determinant independent of a variable	–	Ex 4.1 Q11
7	Cofactor expansion along a specified row	–	Ex 4.1 Q12
II. Properties of Determinants			
8	Det. = 0: identical/proportional rows-columns	Yes	Ex 4.2 Q1(i); Q1(iii) (alt.)
9	$ kA = k^n A $ and related properties	Yes	Ex 4.2 Q7; Q13 (alt.)
10	Solving for x using the row-sum trick	Yes	Ex 4.2 Q15; Q40(i) (alt.)
11	Proving one root via a row operation	–	Ex 4.2 Q16
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Type	Method / Pattern	★	Source(s)
14	Using an A.P. condition with row operations	–	Ex 4.2 Q21
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16	Symmetric/cyclic identities in a, b, c	Yes	Ex 4.2 Q35; Q39 (alt.)
17	Algebraic relation from $\det. = 0$	–	Ex 4.2 Q38
18	Isosceles-triangle proof from $\det. = 0$	–	Ex 4.2 Q42
III. Area of a Triangle			
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20	Checking collinearity of three points	–	Ex 4.3 Q2
21	Unknown coordinate from collinearity	–	Ex 4.3 Q4
22	Equation of a line through two points	–	Ex 4.3 Q5
23	Unknown parameter from a given area	–	Ex 4.3 Q6
IV. Adjoint and Inverse of a Square Matrix			
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25	Singularity: finding unknown for singular A	–	Ex 4.4 Q1(i)
26	Finding A^{-1} using the adjoint formula	Yes	Ex 4.4 Q4(iii); Q21(i) (alt.)
27	Determinant–adjoint relations	–	Ex 4.4 Q5(ii)
28	Verifying $A(\text{adj } A) = A I = (\text{adj } A)A$	–	Ex 4.4 Q11(i)
29	Finding A^{-1} via a matrix polynomial identity	Yes	Ex 4.4 Q23(i); Q24(i) (alt.)
30	Solving a matrix equation using inverses	–	Ex 4.4 Q28(ii)
31	Proving A^{-1} equals a special form of A	–	Ex 4.4 Q25(ii)
V. Systems of Linear Equations			
32	Solving a 2×2 system by matrix method	–	Ex 4.5 Q1(i)
33	Consistency/inconsistency of a system	–	Ex 4.5 Q2(i,ii)
34	Solving a 3×3 system by matrix method	Yes	Ex 4.5 Q9; Q4(ii) (alt.)
35	Reciprocal-substitution systems	–	Ex 4.5 Q5
36	Word problems as linear systems (7 examples total)	Yes	Ex 4.5 Q8, Q7 + 5 original practice
37	Homogeneous systems	–	Ex 4.5 Q15(ii)
38	Condition for a unique solution	–	Ex 4.5 Q6
VI. Higher-Order / Competency-Focused Questions			
39	Higher-degree identity (a^2, b^2, c^2 entries)	Yes	Ex 4.2 Q30 (CFQ)
40	Squared-determinant identity, equilateral triangle	–	Example 7, Sec 4.3 (CFQ)
41	Proving $A^{-1} = A'$ via row dot-products	–	Ex 4.4 Q25(i)

Formula Sheet

Basics

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Value of a 2×2 determinant

Basics

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Expansion of a 3×3 determinant along Row 1

Basics

$$A_{ij} = (-1)^{i+j} M_{ij}$$

Cofactor $A_{ij} = \text{sign} \times \text{minor } M_{ij}$ (minor obtained by deleting row i , column j)

Basics

$$\Delta = \sum_j a_{ij} A_{ij} \quad (\text{any fixed row } i)$$

Cofactor expansion: determinant = sum of (element $\times$ its own cofactor) along any row/column

Properties

$$|kA| = k^n |A|, \quad n = \text{order of } A$$

Multiplying every element of one row/column by k multiplies Δ by k ; multiplying the whole matrix by k multiplies Δ by k^n

Properties

$$|A| = |A^T|$$

A determinant is unchanged under transpose

Properties

$$|AB| = |A| |B|$$

Determinant of a matrix product = product of determinants

Properties

$$R_i \leftrightarrow R_j \Rightarrow \Delta \rightarrow -\Delta$$

Interchanging two rows (or columns) reverses the sign of the determinant

Properties

$$\text{Identical or proportional rows/columns} \Rightarrow \Delta = 0$$

If one row (or column) is a scalar multiple of another, the determinant vanishes

Area/Collinearity

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad (\text{absolute value})$$

Area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

Area/Collinearity

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$$

Three points are collinear $\iff$ the triangle they form has zero area

Adjoint/Inverse

$$\text{adj } A = [A_{ij}]^T$$

Adjoint = transpose of the matrix of cofactors

Adjoint/Inverse

$$A(\text{adj } A) = |A| I = (\text{adj } A)A$$

Key identity linking a matrix, its adjoint and its determinant

Adjoint/Inverse

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad |A| \neq 0$$

A^{-1} exists $\iff A$ is non-singular ($|A| \neq 0$)

Adjoint/Inverse

$$|\text{adj } A| = |A|^{n-1}, \quad n = \text{order of } A$$

Determinant of the adjoint in terms of $|A|$ and order n

Systems of Equations

$$AX = B \Rightarrow X = A^{-1}B, \quad |A| \neq 0$$

Matrix method: unique solution exists iff A is non-singular

Systems of Equations

$$|A| = 0 \text{ and } (adj A)B \neq O \Rightarrow \text{inconsistent}; \quad |A| = 0 \text{ and } (adj A)B = O \Rightarrow \text{consistent, infinite solutions}$$

Test for consistency when A is singular

Systems of Equations

$$AX = O$$

Homogeneous system: always consistent. $|A| \neq 0 \Rightarrow$ only trivial solution $X = O$; $|A| = 0 \Rightarrow$ infinitely many solutions

I. Basics of Determinants

Type 1

A determinant of order 2, with numeric or trigonometric-ratio entries, is asked to be evaluated directly. There is no simplification trick needed – just apply the 2×2 expansion formula and simplify the arithmetic (using standard trig values where needed).

Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Source: Exercise 4.1, Q2(i)

Given: $\begin{vmatrix} \sin 30^\circ & \cos 30^\circ \\ -\sin 60^\circ & \cos 60^\circ \end{vmatrix}$

Solution:

Write down the standard values of the trig ratios involved, since the determinant can only be evaluated once every entry is a number: $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$.

Substitute these into the 2×2 expansion $ad - bc$, taking $a = \sin 30^\circ$, $b = \cos 30^\circ$, $c = -\sin 60^\circ$, $d = \cos 60^\circ$, because that is exactly the pattern of the formula:

$$\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{\sqrt{3}}{2}\right)$$

Simplify each product separately so no sign is lost:

$$= \frac{1}{4} - \left(-\frac{3}{4}\right) = \frac{1}{4} + \frac{3}{4}$$

Add the two fractions (same denominator) to reach the final numeric value:

$$= 1$$

Final Answer

1

Common Mistake: Sign errors while substituting a negative entry (here $-\sin 60^\circ$) into $ad - bc$ – students often drop the negative sign of c and get bc with the wrong sign, flipping the final answer.

Quick Recall: For a 2×2 determinant, just cross-multiply and subtract: top-left $\times$ bottom-right minus top-right $\times$ bottom-left.

Type 2

A 3×3 determinant with purely numeric entries is asked to be evaluated. The method is direct cofactor expansion along the first row (or any convenient row/column with zeros, if present).

Formula Used: $\Delta = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Source: Exercise 4.1, Q9(ii)

Given: $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix}$

Solution:

Expand along Row 1, since it has no zero entries to exploit and expanding along the top row is the most systematic default:

$$\Delta = 3 \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} - (-4) \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} + 5 \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix}$$

Evaluate each 2×2 minor separately, because a mistake in one minor would silently corrupt the whole answer:

$$\begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} = 1 + 6 = 7, \quad \begin{vmatrix} 1 & -2 \\ 2 & 1 \end{vmatrix} = 1 + 4 = 5, \quad \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 3 - 2 = 1$$

Substitute the minors back and simplify, being careful with the $-(-4)$ sign:

$$\Delta = 3(7) + 4(5) + 5(1) = 21 + 20 + 5$$

Final Answer

$$\Delta = 46$$

Common Mistake: Forgetting the alternating sign pattern $+, -, +$ in front of the three minors – especially dropping the sign flip on the middle term.

Quick Recall: Expand along the row/column with the most zeros; otherwise the top row, with the $+, -, +$ sign pattern, and add up the three signed 2×2 minors.

Type 3

A specific element's cofactor is asked for, either directly or via the value of $a_{ij}A_{ij}$. This tests whether the minor is formed correctly (deleting the right row and column) and whether the $(-1)^{i+j}$ sign is applied correctly.

Formula Used: $A_{ij} = (-1)^{i+j} M_{ij}$

Source: Exercise 4.1, Q4(i)

Given: $A = \begin{bmatrix} 5 & 6 & -3 \\ -4 & 3 & 2 \\ -4 & -7 & 3 \end{bmatrix}$, find the cofactor of a_{21}

Solution:

Identify the element: $a_{21} = -4$ sits in Row 2, Column 1, so $i = 2$, $j = 1$.

Form the minor M_{21} by deleting Row 2 and Column 1 from A , since a minor is always the determinant of what remains after deleting that element's own row and column:

$$M_{21} = \begin{vmatrix} 6 & -3 \\ -7 & 3 \end{vmatrix} = (6)(3) - (-3)(-7) = 18 - 21 = -3$$

Attach the sign $(-1)^{i+j} = (-1)^{2+1} = (-1)^3 = -1$, because the cofactor is the signed minor, not the minor itself:

$$A_{21} = (-1)(-3)$$

Final Answer

$$A_{21} = 3$$

Common Mistake: Deleting the wrong row/column (e.g. deleting Column 2 instead of Column 1), or forgetting to attach the negative sign when $i + j$ is odd.

Quick Recall: Cross out the element's row and column, take that 2×2 determinant, then slap on a $+$ or $-$ depending on whether row+column is even or odd.

Type 4

A 2×2 determinant is set equal to a given value (or another determinant), and the resulting equation must be solved for an unknown x . Sometimes the answer must additionally be filtered by a stated domain (such as $x \in \mathbb{N}$ or $x \in \mathbb{I}$), which can eliminate one of the algebraic roots.

★ **Board Favorite** – This is one of the most repeated 1-2 mark question formats in ISC Maths papers on Determinants – setting up and solving a linear or quadratic equation from a determinant condition appears across almost every year's paper in some form. Since I don't have the exact past years for this specific book's question set, I'm flagging it as a standard high-yield pattern for this chapter rather than citing invented years.

Formula Used: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

Source: Exercise 4.1, Q5(i)

Given: $\begin{vmatrix} 2x+5 & 3 \\ 5x+2 & 9 \end{vmatrix} = 0$

Solution:

Expand the determinant using $ad - bc$, because the equation only becomes solvable once it is written as an ordinary algebraic equation in x :

$$(2x + 5)(9) - 3(5x + 2) = 0$$

Expand both products carefully, keeping every term:

$$18x + 45 - 15x - 6 = 0$$

Collect like terms (x -terms together, constants together):

$$3x + 39 = 0$$

Isolate x by moving the constant across and dividing:

$$x = -\frac{39}{3}$$

Final Answer

$$x = -13$$

Common Mistake: Sign errors while expanding $3(5x + 2)$ – students often write $-15x + 6$ instead of $-15x - 6$, flipping the sign of the constant term.

Alternate Board-Style Example

Source: Exercise 4.1, Q7(i)

Given: $x \in \mathbb{N}$ and $\begin{vmatrix} x & 3 \\ 4 & x \end{vmatrix} = \begin{vmatrix} 4 & -3 \\ 0 & 1 \end{vmatrix}$

Solution:

Expand both sides separately, since each is an independent 2×2 determinant:

$$\text{LHS} = x^2 - 12, \quad \text{RHS} = (4)(1) - (-3)(0) = 4$$

Equate LHS and RHS to form the equation in x :

$$x^2 - 12 = 4 \Rightarrow x^2 = 16$$

Solve the quadratic by taking square roots on both sides, remembering both signs at this stage:

$$x = \pm 4$$

Apply the stated domain restriction $x \in \mathbb{N}$ (natural numbers), which is the step students most often skip – this is exactly why the domain condition was given in the question:

$$x = -4 \text{ is rejected since } -4 \notin \mathbb{N}$$

Final Answer

$$x = 4$$

Quick Recall: Expand both sides with $ad - bc$, solve the resulting equation for x , and if a domain ($\mathbb{N}$, $\mathbb{I}$, etc.) is stated, throw out any root that doesn't belong to it.

Type 5

A 3×3 determinant with algebraic entries (often skew-symmetric-looking, with $a, -a, b, -b, c, -c$ pattern) must be shown, by full expansion, to equal a given algebraic expression. Unlike Section II, no row/column property shortcut is used here – it's pure expansion and simplification.

Formula Used: $\Delta = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Source: Exercise 4.1, Q10

Given: Prove that
$$\begin{vmatrix} 1 & a & b \\ -a & 1 & c \\ -b & -c & 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

Solution:

Expand along Row 1, since it's the natural first row and none of its entries is zero here:

$$\Delta = 1 \begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} - a \begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} + b \begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix}$$

Evaluate each 2×2 minor on its own, to avoid mixing up terms:

$$\begin{vmatrix} 1 & c \\ -c & 1 \end{vmatrix} = 1 + c^2, \quad \begin{vmatrix} -a & c \\ -b & 1 \end{vmatrix} = -a + bc, \quad \begin{vmatrix} -a & 1 \\ -b & -c \end{vmatrix} = ac + b$$

Substitute back, distributing the $-a$ and b carefully across their respective brackets:

$$\Delta = (1 + c^2) - a(-a + bc) + b(ac + b) = 1 + c^2 + a^2 - abc + abc + b^2$$

Notice that $-abc$ and $+abc$ cancel exactly – this cancellation is the whole point of the identity, so it confirms the expansion is on the right track:

$$\Delta = 1 + a^2 + b^2 + c^2$$

Final Answer

Hence $\Delta = 1 + a^2 + b^2 + c^2$, as required.

Common Mistake: Losing track of the minus sign in front of the a -term and not distributing it across both parts of $(-a + bc)$, which changes the sign of the abc term and stops it from cancelling.

Quick Recall: Expand straight along Row 1 with the $+, -, +$ pattern; in these skew-looking determinants the abc cross-terms are designed to cancel, so if they don't, re-check a sign.

Type 6

A 3×3 determinant contains a variable (usually inside trig functions), and the question asks to show that the value of the determinant does not actually depend on that variable. The method is to expand fully and watch the variable cancel out, usually via an identity like $\sec^2 x - \tan^2 x = 1$.

Formula Used: $\Delta = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Formula Used: $\sec^2 x - \tan^2 x = 1$

Source: Exercise 4.1, Q11

Given: Show that $\begin{vmatrix} 0 & \tan x & 1 \\ 1 & -\sec x & 0 \\ \sec x & 0 & \tan x \end{vmatrix}$ is independent of x

Solution:

Expand along Row 1, choosing it because it already has a zero entry which simplifies the work:

$$\Delta = 0 \begin{vmatrix} -\sec x & 0 \\ 0 & \tan x \end{vmatrix} - \tan x \begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} + 1 \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix}$$

The first term vanishes automatically because its coefficient is 0, so only the last two minors need evaluating:

$$\begin{vmatrix} 1 & 0 \\ \sec x & \tan x \end{vmatrix} = \tan x, \quad \begin{vmatrix} 1 & -\sec x \\ \sec x & 0 \end{vmatrix} = 0 - (-\sec^2 x) = \sec^2 x$$

Substitute back:

$$\Delta = -\tan x(\tan x) + \sec^2 x = -\tan^2 x + \sec^2 x$$

Apply the trig identity $\sec^2 x - \tan^2 x = 1$, which is exactly the simplification that removes x from the final value – this is the step that proves independence:

$$\Delta = 1$$

Final Answer

$\Delta = 1$ for all x , so the determinant is independent of x .

Common Mistake: Stopping at $\sec^2 x - \tan^2 x$ without recognising/applying the Pythagorean identity, and leaving the "proof" incomplete because the x -dependence looks like it's still there.

Quick Recall: Expand fully, then look for a Pythagorean trig identity (like $\sec^2 - \tan^2 = 1$) to collapse the expression to a pure number.

Type 7

A 3×3 determinant must be evaluated, but the question specifically instructs which row (or column) to expand along – usually to test whether cofactor expansion works correctly along a row other than the first.

Formula Used: $\Delta = \sum_j a_{ij}A_{ij}$ (fixed row i)

Source: Exercise 4.1, Q12

Given: Using cofactors of elements of the second row, evaluate $\Delta = \begin{vmatrix} 5 & 3 & 8 \\ 2 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix}$

Solution:

Identify Row 2's entries, since the question specifically requires expanding along that row: $a_{21} = 2$, $a_{22} = 0$, $a_{23} = 1$.

Find each cofactor by deleting Row 2 and the relevant column, then applying $(-1)^{i+j}$:

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 3 & 8 \\ 2 & 3 \end{vmatrix} = -(9 - 16) = 7$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} 5 & 8 \\ 1 & 3 \end{vmatrix} = (15 - 8) = 7$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} = -(10 - 3) = -7$$

Combine each Row-2 entry with its own cofactor and add, since expansion along a row means $\sum a_{2j}C_{2j}$:

$$\Delta = (2)(7) + (0)(7) + (1)(-7) = 14 + 0 - 7$$

Final Answer

$\Delta = 7$

Common Mistake: Expanding along Row 1 out of habit even though the question specifically asks for Row 2 – this often still gives the right numeric answer (since the determinant's value doesn't depend on the row chosen) but loses marks for not following the instruction/method asked for.

Quick Recall: Whichever row/column you're told to use, find that row's three cofactors (sign, then 2×2 minor), then multiply each entry by its own cofactor and add.

II. Properties of Determinants

Type 8

The question asks to prove, *without expanding*, that a determinant equals zero. On inspection, one row (or column) turns out to be an exact scalar multiple of another row (or column) – either immediately, or after one simple row/column operation reveals it.

★ **Board Favorite** – This "spot the proportional row/column" pattern is one of the most repeated very-short-answer question types on Determinants – it is a standard high-yield pattern for this chapter, appearing in nearly every year's paper in some form (exact years not confirmed from this book, so stated generally rather than invented).

Formula Used: If $R_i = k \cdot R_j$ (or $C_i = k \cdot C_j$) for some scalar k , then $\Delta = 0$

Source: Exercise 4.2, Q1(i)

Given: $\begin{vmatrix} 102 & 18 & 36 \\ 1 & 3 & 4 \\ 17 & 3 & 6 \end{vmatrix}$

Solution:

Look for a relationship between rows before doing any arithmetic – scanning Row 1 and Row 3 (rather than expanding blindly) is what this "without expanding" instruction is testing:

$$R_1 = (102, 18, 36), \quad R_3 = (17, 3, 6)$$

Check whether R_1 is a multiple of R_3 , since $102 = 6 \times 17$, $18 = 6 \times 3$, and $36 = 6 \times 6$ all hold together:

$$R_1 = 6R_3$$

Since one row is exactly a scalar multiple of another row, quote the property directly instead of expanding – that is the entire point of the "without expanding" instruction:

$$\Delta = 0 \quad (\text{because } R_1 \text{ and } R_3 \text{ are proportional})$$

Final Answer

$$\Delta = 0$$

Common Mistake: Reaching for full cofactor expansion anyway (which still gives the right numeric answer but wastes time and loses the "without expanding" method marks), instead of spotting the proportional row.

Alternate Board-Style Example*Source: Exercise 4.2, Q1(iii)*

Given: $\begin{vmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{vmatrix}$

Solution:

No row or column is proportional to another outright here, so instead create one using a column operation – the numbers going up by a constant gap in each row is the clue to try $C_2 \rightarrow C_2 - C_1$:

$$C_2 \rightarrow C_2 - C_1 : (3 - 1, 9 - 7, 15 - 13) = (2, 2, 2)$$

Apply a second, similar operation on the original C_3 , since both new columns being identical is exactly what would force $\Delta = 0$:

$$C_3 \rightarrow C_3 - C_2^{(\text{original})} : (5 - 3, 11 - 9, 17 - 15) = (2, 2, 2)$$

Compare the two new columns – they are now identical:

$$\text{New } C_2 = \text{New } C_3 = (2, 2, 2)$$

Quote the property for identical columns directly:

$$\Delta = 0 \quad (\text{two identical columns after the operation})$$

Final Answer

$$\Delta = 0$$

Quick Recall: Before expanding anything, scan for a row/column that's a multiple of another, or try a simple row/column subtraction to reveal one – if you find it, the determinant is instantly 0.

Type 9

The question gives $|A|$ (or enough information to find it) and asks for the value of a related determinant – such as $|kA|$, $|A^T|$, $|AB|$, or $|A^n|$ – using the standard determinant properties rather than reconstructing and re-evaluating the whole matrix.

★ **Board Favorite** – Very frequently tested as a quick 1-mark objective/VSA question in ISC papers on this chapter – a standard high-yield pattern, since I don't have confirmed specific years for this exact question set.

Formula Used: $|kA| = k^n|A|$, n = order of the square matrix A

Formula Used: $|AB| = |A| |B|$

Source: Exercise 4.2, Q7

Given: A is a square matrix of order 3 and $|A| = 5$, find $|3A|$

Solution:

Identify the order of the matrix, since the power in $|kA| = k^n|A|$ depends on it, not on the scalar 3: here $n = 3$.

Apply the scaling property directly, substituting $k = 3$ and the given $|A| = 5$:

$$|3A| = 3^3|A| = 27 \times 5$$

Final Answer

$$|3A| = 135$$

Common Mistake: Using $|kA| = k|A|$ (as if only one row were scaled) instead of $k^n|A|$ – students forget that multiplying the *whole matrix* by k scales *every* row by k , contributing a factor of k each time.

Alternate Board-Style Example

Source: Exercise 4.2, Q13

Given: $A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -4 \\ 3 & -2 \end{bmatrix}$, find $|AB|$

Solution:

Rather than multiplying the matrices out and then taking a determinant, use the product property directly – it's faster and is exactly what this type is testing:

$$|AB| = |A| |B|$$

Evaluate each determinant separately using $ad - bc$:

$$|A| = (1)(-1) - (2)(3) = -1 - 6 = -7, \quad |B| = (1)(-2) - (-4)(3) = -2 + 12 = 10$$

Multiply the two determinants:

$$|AB| = (-7)(10)$$

Final Answer

$$|AB| = -70$$

Quick Recall: Don't rebuild the matrix – use $|kA| = k^n|A|$ for scalar multiples and $|AB| = |A||B|$ for products, plugging in the order/given values directly.

Type 10

An algebraic 3×3 determinant, usually with a repeated shifting pattern of x 's down the rows, is set to 0 and must be solved for x . The key trick is to add all three rows (or columns) into one row, which creates a common factor that can be pulled out, leaving an easy 2×2 (or simpler) determinant to finish.

★ **Board Favorite** – The " $R_1 \rightarrow R_1 + R_2 + R_3$ to factor, then simplify" trick is one of the single most recurring long-answer techniques in the Determinants chapter across ISC papers – a standard high-yield pattern (specific past years not confirmed from this book).

Formula Used: $R_i \rightarrow R_i + kR_j \Rightarrow \Delta$ unchanged

Source: Exercise 4.2, Q15

Given:
$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0$$

Solution:

Add all three rows into Row 1, because every column then collapses to the same value $3x+a$ – this is exactly the pattern (x repeating with one $+a$ shifting position) that signals this trick:

$$R_1 \rightarrow R_1 + R_2 + R_3 : \quad (3x+a, 3x+a, 3x+a)$$

Factor $(3x+a)$ out of the new Row 1, since all three entries share it:

$$\Delta = (3x+a) \begin{vmatrix} 1 & 1 & 1 \\ x & x+a & x \\ x & x & x+a \end{vmatrix}$$

Simplify the remaining determinant using $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$, which clears out most of Row 1:

$$\begin{vmatrix} 1 & 0 & 0 \\ x & a & 0 \\ x & 0 & a \end{vmatrix} = a^2$$

Substitute back and set the whole product to 0:

$$\Delta = (3x+a) a^2 = 0$$

Since a is a (nonzero) parameter, the only way this product is zero is if the other factor vanishes:

$$3x+a=0$$

Final Answer

$$x = -\frac{a}{3}$$

Common Mistake: Forgetting to also check/mention that $a^2 = 0$ would require $a = 0$ (a degenerate case), and simply assuming $a \neq 0$ without stating it.

Alternate Board-Style Example

Source: Exercise 4.2, Q40(i)

Given:
$$\begin{vmatrix} 3x-8 & 3 & 3 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix} = 0$$

Solution:

Add all three rows into Row 1 again, the same trick, since each column sums to the same value:

$$R_1 \rightarrow R_1 + R_2 + R_3 : \quad (3x-2, 3x-2, 3x-2)$$

Factor $(3x-2)$ out:

$$\Delta = (3x-2) \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3x-8 & 3 \\ 3 & 3 & 3x-8 \end{vmatrix}$$

Apply $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$ to the reduced determinant:

$$\begin{vmatrix} 1 & 0 & 0 \\ 3 & 3x-11 & 0 \\ 3 & 0 & 3x-11 \end{vmatrix} = (3x-11)^2$$

Set the full product to zero and solve each factor:

$$(3x-2)(3x-11)^2 = 0 \Rightarrow x = \frac{2}{3} \text{ or } x = \frac{11}{3} \text{ (repeated)}$$

Final Answer

$$x = \frac{2}{3} \text{ or } \frac{11}{3}$$

Quick Recall: Sum all three rows (or columns) into one; if that produces a common factor, pull it out and finish off the small leftover determinant.

Type 11

The question gives a cubic-type determinant equation (an unknown x appearing added to the diagonal entries) and asks to show that a specific value is *one* root – without solving the whole cubic. The trick is to substitute the claimed root and show the resulting row sums are all zero, which forces $\Delta = 0$ instantly.

Formula Used: If every row (or column) sums to 0, then $\Delta = 0$

Source: Exercise 4.2, Q16

Given: Show that one root of $\begin{vmatrix} x+a & b & c \\ b & x+c & a \\ c & a & x+b \end{vmatrix} = 0$ is $x = -(a+b+c)$

Solution:

Substitute $x = -(a+b+c)$ directly into the diagonal entries, since that's the value being tested – not solved for from scratch:

$$x+a = -(b+c), \quad x+c = -(a+b), \quad x+b = -(a+c)$$

Add up the entries of Row 1 with this substitution, to check whether the row-sum property applies:

$$(x+a) + b + c = -(b+c) + b + c = 0$$

Do the same for Row 2 and Row 3, since the same cancellation happens by the same structure:

$$\text{Row 2 sum} = b + (x+c) + a = 0, \quad \text{Row 3 sum} = c + a + (x+b) = 0$$

Since every row sums to zero, adding all three columns into one column $C_1 \rightarrow C_1 + C_2 + C_3$ produces an all-zero column, and a determinant with a zero column is automatically zero:

$$\Delta = 0 \text{ when } x = -(a+b+c)$$

Final Answer

Hence $x = -(a+b+c)$ is a root of the given equation.

Common Mistake: Trying to solve the full cubic in x instead of the much shorter row-sum check, which wastes time when only one specific root needs verifying.

Quick Recall: To confirm a claimed root, substitute it in and check if every row (or column) sums to zero – that alone proves $\Delta = 0$ without expanding the cubic.

Type 12

Two (or three) determinants, built from the same letters but with rows and columns rearranged, are claimed to be equal. The proof uses only the sign-change rule for row/column interchange and the fact that $|A| = |A^T|$ – never full expansion.

Formula Used: Interchanging two rows or two columns multiplies Δ by -1

Formula Used: $|A| = |A^T|$

Source: Exercise 4.2, Q17

Given: Without expanding, prove
$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} x & y & z \\ p & q & r \\ a & b & c \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$$

Solution:

For the first equality, notice the rows have simply been cycled downward by one position – this can be achieved with two row-interchanges (swap $R_1 \leftrightarrow R_2$, then swap the new $R_2 \leftrightarrow R_3$), since a full 3-row cyclic shift is always two transpositions:

$$2 \text{ interchanges} \Rightarrow \text{sign changes twice} \Rightarrow \text{net sign unchanged}$$

So the first two determinants are equal.

For the second equality, start from the original matrix and take its transpose, since $|A| = |A^T|$ leaves the value unchanged:

$$\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix} \quad (\text{transpose})$$

Now interchange Row 1 and Row 2 of this transposed matrix (one sign change), because that produces the b, y, q row on top, matching the target's row order:

$$\begin{vmatrix} b & y & q \\ a & x & p \\ c & z & r \end{vmatrix} = - \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$$

Then interchange Column 1 and Column 2 (a second sign change), because that's exactly what turns $(b, y, q)/(a, x, p)/(c, z, r)$ into the target column order $(y, b, q)/(x, a, p)/(z, c, r)$:

$$\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix} = - \begin{vmatrix} b & y & q \\ a & x & p \\ c & z & r \end{vmatrix}$$

Two sign changes cancel, so combining both steps recovers the original value.

Final Answer

Hence all three determinants are equal, as required.

Common Mistake: Losing count of how many row/column interchanges were used – an odd number of swaps flips the sign, so it's essential to track parity, not just "rearrange and assume it's fine".

Quick Recall: Rearranging rows/columns never changes a determinant's value as long as you've used an even number of swaps (or you've also transposed, which never changes it at all) – count the swaps to be sure of the sign.

Type 13

An algebraic identity must be proved *using properties*: adding all rows (or all columns) into one creates a common factor across an entire row/column, which is then pulled out, leaving a much simpler determinant to expand and match against the required result.

★ **Board Favorite** – This "sum rows/columns to create a common factor, then expand the remainder" approach is the single most common long-answer proof technique in this exercise – a standard high-yield pattern across ISC long-answer questions on Determinants.

Formula Used: $R_i \rightarrow R_i + kR_j \Rightarrow \Delta$ unchanged

Source: Exercise 4.2, Q24(i)

Given: Prove
$$\begin{vmatrix} 1+a & b & c \\ a & 1+b & c \\ a & b & 1+c \end{vmatrix} = 1+a+b+c$$

Solution:

Add all three columns into Column 1, because every row's three entries happen to sum to the same value $1+a+b+c$ – that repeated total across every row is exactly the signal to try this operation:

$$C_1 \rightarrow C_1 + C_2 + C_3 : \quad \text{each row becomes } (1+a+b+c)$$

Factor $(1+a+b+c)$ out of the new Column 1:

$$\Delta = (1+a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & 1+b & c \\ 1 & b & 1+c \end{vmatrix}$$

Clear out the rest of Column 1 using $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, since Column 1 is now all 1's, making this subtraction simple:

$$\begin{vmatrix} 1 & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Expand this triangular-shaped determinant along Column 1 (only the top entry is nonzero):

$$= 1 \times (1 \times 1 - 0 \times 0) = 1$$

Multiply back the factor pulled out earlier:

$$\Delta = (1 + a + b + c) \times 1$$

Final Answer

$\Delta = 1 + a + b + c$, as required.

Common Mistake: Adding the columns/rows incorrectly (e.g. missing that the diagonal 1's only appear once per row) and getting a wrong common factor, or forgetting to re-multiply the factored term back in at the end.

Alternate Board-Style Example

Source: Exercise 4.2, Q23

Given: Prove
$$\begin{vmatrix} a+b & b+c & c+a \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Solution:

Add all three rows into Row 1 (hint-suggested operation), since every column sums to $2(a + b + c)$:

$$R_1 \rightarrow R_1 + R_2 + R_3 : (2(a + b + c), 2(a + b + c), 2(a + b + c))$$

Factor $2(a + b + c)$ out of the new Row 1:

$$\Delta_{LHS} = 2(a + b + c) \begin{vmatrix} 1 & 1 & 1 \\ b+c & c+a & a+b \\ c+a & a+b & b+c \end{vmatrix}$$

Simplify with $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$ on the reduced determinant, to clear Row 1 to $(1, 0, 0)$:

$$\begin{vmatrix} 1 & 0 & 0 \\ b+c & a-b & a-c \\ c+a & b-c & b-a \end{vmatrix}$$

Expand along Row 1 and simplify the remaining 2×2 determinant:

$$(a - b)(b - a) - (a - c)(b - c) = -a^2 - b^2 - c^2 + ab + bc + ca$$

So $\Delta_{LHS} = 2(a + b + c)(-a^2 - b^2 - c^2 + ab + bc + ca) = -2(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.

Now use the standard algebraic identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ to rewrite this as

$$\Delta_{LHS} = -2(a^3 + b^3 + c^3 - 3abc) = 2(3abc - a^3 - b^3 - c^3)$$

Directly expand the RHS determinant $|a \ b \ c; \ b \ c \ a; \ c \ a \ b|$ along Row 1 to check it matches:

$$= a(cb - a^2) - b(b^2 - ac) + c(ab - c^2) = 3abc - a^3 - b^3 - c^3$$

Multiplying by 2 (as required by the RHS of the identity) gives exactly Δ_{LHS} .

Final Answer

Hence $\Delta_{LHS} = 2 \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, as required.

Quick Recall: When every row (or column) adds up to the same total, sum them into one row/column first – that’s almost always the fastest route to a clean common factor.

Type 14

The question states that certain quantities (a, b, c , or similar) are in Arithmetic Progression, and asks to evaluate or prove something about a determinant built from linear expressions in those quantities and a variable. The A.P. condition ($2b = a + c$, or similar) is used inside a row operation to force a zero row.

Formula Used: $2b = a + c$ (condition for a, b, c in A.P.)

Source: Exercise 4.2, Q21

Given: a, b, c are in A.P.; find $\begin{vmatrix} 2y + 4 & 5y + 7 & 8y + a \\ 3y + 5 & 6y + 8 & 9y + b \\ 4y + 6 & 7y + 9 & 10y + c \end{vmatrix}$

Solution:

Notice the row coefficients follow a pattern (each entry in Row 3 is "Row 1 + $2 \times$ common difference of Row 2 vs Row 1"), which suggests trying $R_1 \rightarrow R_1 - 2R_2 + R_3$ to cancel the y -terms completely:

$$\text{Column-wise: } (2y + 4) - 2(3y + 5) + (4y + 6) = 0, \quad (5y + 7) - 2(6y + 8) + (7y + 9) = 0$$

Do the same for the third column, where the A.P. condition on a, b, c finally comes into play:

$$(8y + a) - 2(9y + b) + (10y + c) = (8y - 18y + 10y) + (a - 2b + c) = 0 + (a + c - 2b)$$

Apply the given condition that a, b, c are in A.P., i.e. $a + c = 2b$, which makes this last entry vanish too:

$$a + c - 2b = 0$$

So the new Row 1 is entirely $(0, 0, 0)$, and a determinant with a zero row is immediately zero:

$$\Delta = 0$$

Final Answer

$$\Delta = 0$$

Common Mistake: Trying to expand the determinant directly with the messy linear entries instead of spotting the row combination that uses the A.P. condition – this leads to a long, error-prone calculation for no reason.

Quick Recall: When a chapter condition like "in A.P." is given alongside a determinant, look for a row/column combination that turns that condition into a zero row – it's almost always $R_1 - 2R_2 + R_3$ or similar.

Type 15

A determinant with entries like x, x^2, yz (or purely x, x^2, x^3 -type powers) must be shown to equal a product of differences, such as $(x - y)(y - z)(z - x)$. These are Vandermonde-style determinants: the method is to convert the mixed-product entries into pure power-columns (using row/column scaling), then use the standard $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$ reduction.

★ **Board Favorite** – Vandermonde-type "prove equals a product of differences" questions are a classic, frequently-tested long-answer pattern in this chapter – a standard high-yield pattern (exact past years not confirmed from this book).

Formula Used: $R_i \rightarrow kR_i \Rightarrow \Delta \rightarrow k\Delta$; a common factor in a row/column can be taken out

Source: Exercise 4.2, Q26(i)

Given: Prove
$$\begin{vmatrix} 1 & x & yz \\ 1 & y & zx \\ 1 & z & xy \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (x - y)(y - z)(z - x)$$

Solution:

Multiply Row 1 by x , Row 2 by y , and Row 3 by z , since this turns each yz, zx, xy entry into the same product xyz – this scales the whole determinant by the product xyz , which will be cancelled later:

$$xyz \cdot \Delta = \begin{vmatrix} x & x^2 & xyz \\ y & y^2 & xyz \\ z & z^2 & xyz \end{vmatrix}$$

Factor xyz out of the now-common Column 3:

$$xyz \cdot \Delta = xyz \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

Cancel xyz from both sides (valid since $x, y, z \neq 0$ for this identity to be meaningful):

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix}$$

Cyclically shift the columns of this determinant (move Column 3 to the front) – a 3-column cyclic shift is two transpositions, an even number, so the value is unchanged:

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

This proves the first equality. Now evaluate this standard Vandermonde determinant using $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, which is the standard way to create zeros in the first column:

$$\begin{vmatrix} 1 & x & x^2 \\ 0 & y - x & y^2 - x^2 \\ 0 & z - x & z^2 - x^2 \end{vmatrix}$$

Expand along Column 1, and factor each entry using $y^2 - x^2 = (y - x)(y + x)$:

$$= (y - x)(z^2 - x^2) - (y^2 - x^2)(z - x) = (y - x)(z - x)[(z + x) - (y + x)]$$

Simplify the bracket and rearrange the sign of each factor to match the required form:

$$= (y - x)(z - x)(z - y) = (x - y)(y - z)(z - x)$$

Final Answer

Hence both determinants equal $(x - y)(y - z)(z - x)$.

Common Mistake: Trying to expand the original mixed-entry determinant directly instead of first converting it to pure powers – this leads to unmanageable algebra with xy, yz, zx terms.

Alternate Board-Style Example

Source: Exercise 4.2, Q31

Given: Prove $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} = (a - b)(b - c)(c - a)(ab + bc + ca)$

Solution:

Scale Column 1 by a , Column 2 by b , Column 3 by c – chosen because this turns $bc \cdot a$, $ca \cdot b$, $ab \cdot c$ into the same value abc in Row 3, at the cost of an overall factor abc :

$$abc \cdot \Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ abc & abc & abc \end{vmatrix}$$

Factor abc from the now-common Row 3, and cancel it from both sides:

$$\Delta = \begin{vmatrix} a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \\ 1 & 1 & 1 \end{vmatrix}$$

Cyclically move Row 3 to the top (two row swaps, even, unchanged), then transpose (also unchanged), which together give exactly the required middle form:

$$\Delta = \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

This proves the first equality. Reduce this with $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$ and expand along Column 1, factoring $(b - a)$ and $(c - a)$ out of Row 2 and Row 3 respectively, then simplifying the remaining bracket:

$$\Delta = (b - a)(c - a)(c - b)(ab + bc + ca)$$

Rearranging the sign of each bracketed difference (each swap of order flips one sign, and three swaps give back a net positive) shows this equals the required form:

$$\Delta = (a - b)(b - c)(c - a)(ab + bc + ca)$$

Final Answer

Hence $\Delta = (a - b)(b - c)(c - a)(ab + bc + ca)$, as required.

Quick Recall: Whenever a determinant mixes plain letters with their squares/cubes or products, try scaling rows/columns to make one row or column uniform first – that’s the doorway into the standard Vandermonde reduction.

Type 16

A more elaborate identity, symmetric or cyclic in a, b, c (or x, y, z), must be proved using a combination of row and column operations – often a clever substitution such as letting $s = x + y + z$ helps spot a common column before the standard $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$ reduction.

★ **Board Favorite** – The long-answer “prove this cyclic/symmetric identity using properties” question is asked almost every year in some form in the Determinants chapter – a standard high-yield pattern for board exams (exact years not confirmed from this book).

Formula Used: $R_i \rightarrow R_i + kR_j \Rightarrow \Delta$ unchanged

Source: Exercise 4.2, Q35

Given:
$$\begin{vmatrix} y+z & x+y & x \\ z+x & y+z & y \\ x+y & z+x & z \end{vmatrix} = x^3 + y^3 + z^3 - 3xyz$$

Solution:

Let $s = x + y + z$ to rewrite every entry compactly, since $y + z = s - x$, $z + x = s - y$, $x + y = s - z$ – this substitution is what reveals the hidden common column:

$$\begin{vmatrix} s-x & s-z & x \\ s-y & s-x & y \\ s-z & s-y & z \end{vmatrix}$$

Add Column 3 to Column 1 ($C_1 \rightarrow C_1 + C_3$), because $(s-x) + x = (s-y) + y = (s-z) + z = s$ in every row – exactly the common value this substitution was designed to expose:

$$\begin{vmatrix} s & s-z & x \\ s & s-x & y \\ s & s-y & z \end{vmatrix}$$

Factor s out of the now-common Column 1:

$$\Delta = s \begin{vmatrix} 1 & s-z & x \\ 1 & s-x & y \\ 1 & s-y & z \end{vmatrix}$$

Clear Column 1 with $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$, since Column 1 is now all 1's:

$$\begin{vmatrix} 1 & s-z & x \\ 0 & z-x & y-x \\ 0 & z-y & z-x \end{vmatrix}$$

Expand along Column 1 and simplify the remaining 2×2 bracket:

$$= (z-x)^2 - (y-x)(z-y) = x^2 + y^2 + z^2 - xy - yz - zx$$

Multiply back the factor $s = x + y + z$ pulled out earlier, and use the standard identity $(x+y+z)(x^2+y^2+z^2-xy-yz-zx) = x^3+y^3+z^3-3xyz$:

$$\Delta = (x+y+z)(x^2+y^2+z^2-xy-yz-zx) = x^3+y^3+z^3-3xyz$$

Final Answer

Hence $\Delta = x^3 + y^3 + z^3 - 3xyz$, as required.

Common Mistake: Trying to expand this determinant directly in terms of x, y, z without first substituting $s = x + y + z$ – the algebra becomes far messier and error-prone without spotting the hidden common column.

Alternate Board-Style Example

Source: Exercise 4.2, Q39

Given: Without expanding, prove $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$

Solution:

Expand each squared bracket symbolically (without touching the first row) to see the underlying structure: $(a+1)^2 = a^2 + 2a + 1$ and $(a-1)^2 = a^2 - 2a + 1$, so Row 2 and Row 3 are Row 1 shifted by $\pm 2a + 1$ (and similarly for b, c).

Subtract Row 1 from Row 2 and from Row 3, since that's exactly what strips away the shared a^2, b^2, c^2 part:

$$R_2 \rightarrow R_2 - R_1 : (2a + 1, 2b + 1, 2c + 1), \quad R_3 \rightarrow R_3 - R_1 : (-2a + 1, -2b + 1, -2c + 1)$$

Add the new Row 3 to the new Row 2 ($R_3 \rightarrow R_3 + R_2$), because the $\pm 2a$ terms cancel, leaving a constant row:

$$R_3 \rightarrow R_3 + R_2 : (2, 2, 2)$$

Factor 2 out of the new Row 3:

$$\Delta = 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ 2a + 1 & 2b + 1 & 2c + 1 \\ 1 & 1 & 1 \end{vmatrix}$$

Subtract the (now-simplified) Row 3 from Row 2 ($R_2 \rightarrow R_2 - R_3$), which removes the $+1$ and leaves a pure multiple of a, b, c :

$$R_2 \rightarrow R_2 - R_3 : (2a, 2b, 2c)$$

Factor 2 out of this row as well:

$$\Delta = 2 \times 2 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix} = 4 \begin{vmatrix} a^2 & b^2 & c^2 \\ a & b & c \\ 1 & 1 & 1 \end{vmatrix}$$

Final Answer

Hence the identity is proved, with an overall factor of 4 emerging from two separate row operations.

Quick Recall: For symmetric identities in a, b, c , hunt for row/column combinations that cancel the "extra" terms first (like the $\pm 2a$ here) – the common factor almost always appears in two stages, not one.

Type 17

A determinant involving unknowns a, b, c (all nonzero, or otherwise constrained) is given equal to zero, and the question asks to derive an algebraic relation between a, b, c – not a numeric value. The method is to expand the determinant as normal, set the result to zero, and then algebraically massage that equation (often by dividing through by abc) into the required form.

Formula Used: $\Delta = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Source: Exercise 4.2, Q38

Given: a, b, c are all non-zero, and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$; prove $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$

Solution:

Expand the determinant along Row 1 in the ordinary way, since there's no shortcut property that avoids expansion here – the relation being asked for only appears after full expansion:

$$(1+a)[(1+b)(1+c) - 1] - 1[(1+c) - 1] + 1[1 - (1+b)]$$

Simplify each bracket in turn: $(1+b)(1+c) - 1 = b + c + bc$, the middle bracket is c , and the last bracket is $-b$:

$$(1+a)(b+c+bc) - c - b$$

Expand the remaining product carefully:

$$= b + c + bc + ab + ac + abc - c - b = ab + bc + ca + abc$$

Set this expansion equal to zero, as given:

$$ab + bc + ca + abc = 0$$

Divide the whole equation by abc (valid since a, b, c are all nonzero, exactly as stated in the question), because that's what turns products into the required reciprocal form:

$$\frac{ab}{abc} + \frac{bc}{abc} + \frac{ca}{abc} + \frac{abc}{abc} = 0 \Rightarrow \frac{1}{c} + \frac{1}{a} + \frac{1}{b} + 1 = 0$$

Final Answer

Hence $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + 1 = 0$, as required.

Common Mistake: Dividing by abc without first checking/mentioning that $a, b, c \neq 0$ – the question states this, but it's easy to forget to justify the division step explicitly, which is often a method mark.

Quick Recall: Expand fully, set the result to zero, then divide through by whatever product turns the equation into the required reciprocal (or other algebraic) form.

Type 18

A determinant built from trigonometric ratios of the angles of a triangle ($\cos A, \cos B, \cos C$) is given equal to zero, and the question asks to prove the triangle is isosceles. The method reduces the determinant to a product of differences like $(\cos B - \cos A)(\cos C - \cos A)(\cos B - \cos C)$, and setting this to zero forces two angles to be equal.

Formula Used: If $\cos \theta_1 = \cos \theta_2$ for angles of a triangle ($0 < \theta < \pi$), then $\theta_1 = \theta_2$

Source: Exercise 4.2, Q42

Given: In $\triangle ABC$,
$$\begin{vmatrix} 1 & 1 & 1 \\ 1 + \cos A & 1 + \cos B & 1 + \cos C \\ \cos A + \cos^2 A & \cos B + \cos^2 B & \cos C + \cos^2 C \end{vmatrix} = 0$$

Solution:

Write $p = \cos A$, $q = \cos B$, $r = \cos C$ to keep the algebra manageable, since the structure of the proof doesn't depend on these being angles until the very last step.

Apply $C_2 \rightarrow C_2 - C_1$, $C_3 \rightarrow C_3 - C_1$, which is the natural move since Row 1 is already all 1's (subtracting will create zeros there):

$$\text{Row 1: } (1, 0, 0)$$

For Row 2, the same operation gives $(1 + p, q - p, r - p)$; for Row 3, use $C_2 - C_1 = (q + q^2) - (p + p^2) = (q - p)(1 + p + q)$ (factoring the difference of squares), and similarly $C_3 - C_1 = (r - p)(1 + p + r)$ for Row 3, because expressing each new entry with the common factor $(q - p)$ or $(r - p)$ is what makes the next expansion clean:

$$\begin{vmatrix} 1 & 0 & 0 \\ 1 + p & q - p & r - p \\ p + p^2 & (q - p)(1 + p + q) & (r - p)(1 + p + r) \end{vmatrix}$$

Expand along Row 1, then factor $(q - p)$ and $(r - p)$ out of the resulting 2×2 determinant:

$$(q - p)(r - p)(1 + p + q) - (r - p)(q - p)(1 + p + r) = (q - p)(r - p)[(1 + p + q) - (1 + p + r)]$$

Simplify the bracket, which collapses to just $q - r$:

$$\Delta = (q - p)(r - p)(q - r)$$

Set this product to zero, as given in the question:

$$(q - p)(r - p)(q - r) = 0 \Rightarrow q = p, \text{ or } r = p, \text{ or } q = r$$

Translate back into the triangle's angles, using that cosine is one-to-one for angles strictly between 0 and π (the range of a triangle's interior angles) – this is the step that turns an algebraic equality into an angle equality:

$$\cos B = \cos A, \text{ or } \cos C = \cos A, \text{ or } \cos B = \cos C \Rightarrow B = A, \text{ or } C = A, \text{ or } B = C$$

Final Answer

At least two angles of $\triangle ABC$ are equal, so the triangle is isosceles.

Common Mistake: Stopping at the algebraic result $(q-p)(r-p)(q-r) = 0$ without explicitly translating it back to angle equality via the injectivity of cosine on $(0, \pi)$ – the geometric conclusion needs that last justifying sentence.

Quick Recall: Row-reduce to expose a product of differences in $\cos A, \cos B, \cos C$; once that product is zero, two of the angles must be equal, which is exactly what "isosceles" means.

III. Area of a Triangle using Determinants

Type 19

Three vertices of a triangle are given as coordinates, and the area must be found using the determinant formula. Since area is always positive, the absolute value of the determinant expression is taken at the end.

★ **Board Favorite** – The determinant-based area formula is tested almost every year in some form (direct area, or reversed to find an unknown) – a standard high-yield pattern for this chapter (exact past years not confirmed from this book).

Formula Used: $\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ (taking the absolute value)

Source: Exercise 4.3, Q1(i)

Given: Vertices $(3, 8)$, $(-4, 2)$, $(5, 1)$

Solution:

Set up the determinant with each vertex as a row, appending a 1 in the third column – this is the fixed template the area formula requires:

$$\begin{vmatrix} 3 & 8 & 1 \\ -4 & 2 & 1 \\ 5 & 1 & 1 \end{vmatrix}$$

Expand along Row 1, since none of its entries is zero and there's no special structure to exploit:

$$3(2 \times 1 - 1 \times 1) - 8(-4 \times 1 - 1 \times 5) + 1(-4 \times 1 - 2 \times 5)$$

Simplify each bracket carefully, since a sign slip in any one of the three brackets changes the final area:

$$= 3(1) - 8(-9) + (-14) = 3 + 72 - 14$$

Take half of this value, since the area formula has a leading $\frac{1}{2}$:

$$\text{Area} = \frac{1}{2}(61)$$

Final Answer

Area = 30.5 sq. units

Common Mistake: Forgetting the leading $\frac{1}{2}$ in the area formula, or reporting a negative area instead of taking the absolute value if the raw determinant comes out negative.

Alternate Board-Style Example

Source: Exercise 4.3, Q1(ii)

Given: Vertices (2, 7), (1, 1), (10, 8)

Solution:

Set up the same determinant template with these vertices:

$$\begin{vmatrix} 2 & 7 & 1 \\ 1 & 1 & 1 \\ 10 & 8 & 1 \end{vmatrix}$$

Expand along Row 1:

$$2(1 \times 1 - 1 \times 8) - 7(1 \times 1 - 1 \times 10) + 1(1 \times 8 - 1 \times 10)$$

Simplify each bracket:

$$= 2(-7) - 7(-9) + (-2) = -14 + 63 - 2$$

Halve the result for the area:

$$\text{Area} = \frac{1}{2}(47)$$

Final Answer

Area = 23.5 sq. units

Quick Recall: Stack the three vertices as rows with a 1 in the last column, take the determinant, halve it, and drop any negative sign – area can never be negative.

Type 20

Three points are given, and the question asks to check (or prove) whether they are collinear. Since three collinear points form a "triangle" with zero area, the test is simply whether the area determinant evaluates to zero.

Formula Used: Points collinear $\iff \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Source: Exercise 4.3, Q2

Given: Show that $(1, 0)$, $(6, 0)$, $(0, 0)$ are collinear

Solution:

Set up the same determinant template used for area, since collinearity is just the "zero area" special case:

$$\begin{vmatrix} 1 & 0 & 1 \\ 6 & 0 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

Notice that the entire middle column (the y -coordinates) is zero – every point lies on the x -axis – so quote the zero-column property instead of expanding term by term:

$$\text{Column 2 is entirely 0} \Rightarrow \Delta = 0$$

Final Answer

Since the determinant is 0, the three points are collinear.

Common Mistake: Expanding the full determinant by cofactors when a zero row or column is already visible – always scan for a zero row/column first, since it makes the answer immediate.

Quick Recall: Collinear points always make the area determinant exactly zero – set it up the same way as the area formula and check if it collapses to zero.

Type 21

Three points are given, one of which contains an unknown coordinate, and the points are stated (or required) to be collinear. Setting the area determinant to zero gives a linear equation in the unknown.

Formula Used: $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

Source: Exercise 4.3, Q4

Given: Find λ so that $(\lambda, 7)$, $(1, -5)$, $(-4, 5)$ are collinear

Solution:

Set up the collinearity determinant with the unknown λ in place:

$$\begin{vmatrix} \lambda & 7 & 1 \\ 1 & -5 & 1 \\ -4 & 5 & 1 \end{vmatrix} = 0$$

Expand along Row 1, keeping λ as a symbol throughout:

$$\lambda(-5 \times 1 - 1 \times 5) - 7(1 \times 1 - 1 \times (-4)) + 1(1 \times 5 - (-5) \times (-4))$$

Simplify each bracket:

$$= \lambda(-10) - 7(5) + (5 - 20) = -10\lambda - 35 - 15$$

Set the whole expression to zero and solve the resulting linear equation:

$$-10\lambda - 50 = 0$$

Final Answer

$$\lambda = -5$$

Common Mistake: Mixing up which coordinate is x and which is y when placing the unknown in the determinant – always keep (x, y) order consistent with how the formula is set up.

Quick Recall: Plug the unknown into the same 3-row determinant template used for area, set it to zero, and solve the resulting linear equation.

Type 22

Two fixed points are given, and the question asks for the equation of the line through them – using determinants rather than the usual two-point line formula. A general point (x, y) is placed as the third "vertex", and the collinearity condition (area = 0) with the two given points produces the line's equation directly.

Formula Used:
$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Source: Exercise 4.3, Q5

Given: Find the equation of the line through $(1, 2)$ and $(3, 6)$

Solution:

Treat (x, y) as a general point on the required line, and set up the collinearity condition between (x, y) and the two given points – since any point on the line must be collinear with both fixed points:

$$\begin{vmatrix} x & y & 1 \\ 1 & 2 & 1 \\ 3 & 6 & 1 \end{vmatrix} = 0$$

Expand along Row 1, keeping x and y as symbols:

$$x(2 \times 1 - 1 \times 6) - y(1 \times 1 - 1 \times 3) + 1(1 \times 6 - 2 \times 3)$$

Simplify each bracket:

$$= x(-4) - y(-2) + (6 - 6) = -4x + 2y + 0$$

Set the expression to zero, since this is the equation of the line itself:

$$-4x + 2y = 0 \Rightarrow 2y = 4x$$

Final Answer

$$y = 2x$$

Common Mistake: Forgetting that the final bracket (the constant term) can genuinely simplify to zero – students sometimes assume they’ve made an error and abandon the correct working when the line happens to pass through the origin.

Quick Recall: Put a general point (x, y) as the first row above the two fixed points, set the determinant to zero, and simplify – that equation *is* the line.

Type 23

The area of a triangle is given as a fixed number, and one (or two) vertex coordinates contain an unknown. Setting up the area formula and equating it to the given area produces an equation (often with a modulus, since area is non-negative) that must be solved for the unknown.

Formula Used: Area = $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Source: Exercise 4.3, Q6

Given: Find k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$, $(0, 2)$ is 4 sq. units

Solution:

Set up the area determinant with the unknown k in place:

$$\begin{vmatrix} k & 0 & 1 \\ 4 & 0 & 1 \\ 0 & 2 & 1 \end{vmatrix}$$

Expand along Row 1:

$$k(0 \times 1 - 1 \times 2) - 0(4 \times 1 - 1 \times 0) + 1(4 \times 2 - 0 \times 0)$$

Simplify:

$$= k(-2) - 0 + 8 = -2k + 8$$

Set $\frac{1}{2}|-2k + 8|$ equal to the given area of 4, since area must be taken as an absolute value – this is exactly why a $k = 8$ solution (negative raw determinant) is not automatically discarded here:

$$\frac{1}{2}|-2k + 8| = 4 \Rightarrow |4 - k| = 4$$

Split the modulus equation into its two cases, since both branches give valid geometric triangles:

$$4 - k = 4 \Rightarrow k = 0, \quad \text{or} \quad 4 - k = -4 \Rightarrow k = 8$$

Final Answer

$$k = 0 \text{ or } k = 8$$

Common Mistake: Solving only $4 - k = 4$ and stopping, forgetting that a modulus equation $|A| = c$ always has two cases, $A = c$ and $A = -c$ – this drops one of the two valid values of k .

Quick Recall: Set up the area determinant with the unknown in place, equate $\frac{1}{2}$ times its absolute value to the given area, and remember to solve *both* branches of the resulting modulus equation.

IV. Adjoint and Inverse of a Square Matrix

Type 24

The question directly asks for the adjoint of a given matrix. For a 2×2 matrix this has a quick shortcut; the general method (needed for 3×3) is to find every cofactor and then transpose the matrix of cofactors.

Formula Used: $\text{adj } A = [A_{ij}]^T$ (transpose of the cofactor matrix)

Formula Used: For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$: $\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Source: Exercise 4.4, Q3(i)

Given: $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$, find $\text{adj } A$

Solution:

Use the 2×2 shortcut directly: swap the two diagonal entries, and change the sign of the two off-diagonal entries – this is exactly what computing all four cofactors and transposing them reduces to for a 2×2 matrix:

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 3 & -(-1) \\ -4 & 2 \end{bmatrix}$$

Final Answer

$$\text{adj } A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$$

Common Mistake: Swapping only the diagonal but forgetting to negate *both* off-diagonal entries (or negating only one of them).

Quick Recall: For a 2×2 matrix: swap the diagonal, flip the sign of the other two entries – that's the whole adjoint.

Type 25

A matrix contains an unknown (x or k), and the question asks for the value that makes the matrix singular (i.e. non-invertible). Since singular means $|A| = 0$, the method is to expand the determinant, set it to zero, and solve.

Formula Used: A is singular $\iff |A| = 0$

Source: Exercise 4.4, Q1(i)

Given: For what value of x is $\begin{bmatrix} 3-2x & x+1 \\ 2 & 4 \end{bmatrix}$ singular?

Solution:

Write down the determinant of the given matrix, since singularity is defined purely in terms of the determinant:

$$|A| = (3 - 2x)(4) - (x + 1)(2)$$

Expand both products:

$$= 12 - 8x - 2x - 2 = 10 - 10x$$

Set the determinant to zero, because that is the exact condition for singularity:

$$10 - 10x = 0$$

Final Answer

$$x = 1$$

Common Mistake: Confusing "singular" with "non-singular" and accidentally solving $|A| \neq 0$ style reasoning, or making a sign slip while expanding $(x + 1)(2)$.

Quick Recall: "Singular" always means determinant = 0 – write the determinant, set it to zero, solve for the unknown.

Type 26

A specific matrix is given (order 2 or 3), and its inverse must be found using the adjoint formula: compute the determinant, compute the adjoint, then divide the adjoint by the determinant.

★ **Board Favorite** – Finding A^{-1} via the adjoint formula is one of the most directly-tested skills in this chapter, forming the backbone of the later "solve a system of equations" questions too – a standard high-yield pattern (exact past years not confirmed from this book).

Formula Used: $A^{-1} = \frac{1}{|A|} \text{adj } A, \quad |A| \neq 0$

Source: Exercise 4.4, Q4(iii)

Given: Find the inverse of $\begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}$

Solution:

Compute the determinant first, since A^{-1} only exists (and only makes sense to compute) if this is nonzero:

$$|A| = (3)(2) - (-1)(1) = 6 + 1 = 7$$

Find the adjoint using the 2×2 shortcut (swap diagonal, negate off-diagonal):

$$\text{adj } A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Divide the adjoint by the determinant, as the formula requires:

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Final Answer

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$$

Common Mistake: Forgetting to divide by $|A|$ at the end and submitting the adjoint itself as the answer – the adjoint is only the numerator of A^{-1} .

Alternate Board-Style Example

Source: Exercise 4.4, Q21(i)

Given: $A = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 5 & 2 & -1 \end{bmatrix}$, find A^{-1} and verify $A^{-1}A = I_3$

Solution:

Compute $|A|$ by expanding along Row 1, which is efficient here because two of its entries are already zero:

$$|A| = 1[(3)(-1) - (0)(2)] = -3$$

Find every cofactor A_{ij} systematically (deleting the relevant row and column each time, then applying the sign $(-1)^{i+j}$):

$$A_{11} = -3, A_{12} = 3, A_{13} = -9, A_{21} = 0, A_{22} = -1, A_{23} = -2, A_{31} = 0, A_{32} = 0, A_{33} = 3$$

Transpose the cofactor matrix to get the adjoint – rows become columns, which is what makes the adjoint construction easy to get wrong if rushed:

$$\text{adj } A = \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix}$$

Divide by $|A| = -3$ to get A^{-1} :

$$A^{-1} = \frac{1}{-3} \begin{bmatrix} -3 & 0 & 0 \\ 3 & -1 & 0 \\ -9 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & \frac{1}{3} & 0 \\ 3 & \frac{2}{3} & -1 \end{bmatrix}$$

Verify by multiplying $A^{-1}A$ row by row – this check is what confirms no arithmetic slip happened anywhere above:

$$A^{-1}A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \checkmark$$

Final Answer

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & \frac{1}{3} & 0 \\ 3 & \frac{2}{3} & -1 \end{bmatrix}$$

Quick Recall: Three steps, always in this order: find $|A|$, build $\text{adj } A$ from all the cofactors (transposed), then divide every entry of the adjoint by $|A|$.

Type 27

The question gives $|A|$ (and the order of A) and asks for a related quantity built from the adjoint – such as $|\text{adj } A|$ – using the standard determinant-adjoint identities, without ever constructing the actual matrix.

Formula Used: $|\text{adj } A| = |A|^{n-1}$, n = order of A

Source: Exercise 4.4, Q5(ii)

Given: A is a square matrix of order 3 and $|A| = 5$, find $|\text{adj } A|$

Solution:

Identify the order $n = 3$, since the power in the identity depends on it.

Apply the identity directly, substituting $|A| = 5$ and $n = 3$:

$$|\text{adj } A| = |A|^{n-1} = 5^{3-1} = 5^2$$

Final Answer

$$|\text{adj } A| = 25$$

Common Mistake: Using the exponent n instead of $n - 1$ – this identity is easy to misremember, so it's worth re-deriving mentally from $A(\text{adj } A) = |A|I$ before plugging in numbers.

Quick Recall: $|\text{adj } A|$ is $|A|$ raised to (order -1), not to the order itself – for a 3×3 matrix that's $|A|$ squared.

Type 28

The question gives a specific matrix and asks to explicitly verify the identity $A(\text{adj } A) = |A|I = (\text{adj } A)A$ by computing both matrix products and checking they match the scalar multiple of the identity matrix.

Formula Used: $A(\text{adj } A) = |A|I = (\text{adj } A)A$

Source: Exercise 4.4, Q11(i)

Given: $A = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$, verify $A(\text{adj } A) = |A|I_2 = (\text{adj } A)A$

Solution:

Compute $|A|$ first, since it's the target value both products must reduce to (as a multiple of I):

$$|A| = (-2)(4) - (1)(3) = -8 - 3 = -11$$

Find the adjoint using the 2×2 shortcut:

$$\text{adj } A = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix}$$

Multiply $A \cdot (\text{adj } A)$ directly, row by column, to check the claimed identity:

$$A(\text{adj } A) = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} = \begin{bmatrix} -8-3 & 2-2 \\ 12-12 & -3-8 \end{bmatrix} = \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}$$

Compare this to $|A|I_2 = -11 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, which matches exactly – confirming the first half of the identity.

Multiply the other order, $(\text{adj } A) \cdot A$, to check the second half holds too (matrix multiplication isn't generally commutative, so this genuinely needs its own check):

$$(\text{adj } A)A = \begin{bmatrix} 4 & -1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -8-3 & 4-4 \\ 6-6 & -3-8 \end{bmatrix} = \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}$$

Final Answer

$$A(\text{adj } A) = |A|I_2 = (\text{adj } A)A = \begin{bmatrix} -11 & 0 \\ 0 & -11 \end{bmatrix}, \text{ verified.}$$

Common Mistake: Checking only $A(\text{adj } A)$ and assuming $(\text{adj } A)A$ must automatically match – both products should be shown explicitly since matrix multiplication order matters in general.

Quick Recall: Multiply A and its adjoint both ways – both products should collapse to $|A|$ times the identity matrix.

Type 29

The matrix A is shown to satisfy a polynomial equation of its own (e.g. $A^2 - 3A - 7I = O$), reminiscent of the Cayley–Hamilton theorem. Once that identity is established, it can be rearranged algebraically to express A^{-1} directly – without ever computing the adjoint.

★ **Board Favorite** – This "matrix satisfies its own equation, hence find A^{-1} " technique is a favourite long-answer/Competency-Focused pattern in recent ISC papers on this chapter – a standard high-yield pattern (exact years not confirmed from this book).

Formula Used: If $A^2 - pA - qI = O$ ($q \neq 0$), then $A^{-1} = \frac{1}{q}(A - pI)$

Source: Exercise 4.4, Q23(i)

Given: $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$; show $A^2 - 3A - 7I = O$, hence find A^{-1}

Solution:

Compute A^2 by multiplying A by itself, since the whole identity hinges on this being correct:

$$A^2 = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 25 - 3 & 15 - 6 \\ -5 + 2 & -3 + 4 \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

Compute $3A$ for the subtraction:

$$3A = \begin{bmatrix} 15 & 9 \\ -3 & -6 \end{bmatrix}$$

Subtract $3A$ from A^2 , then subtract $7I$, to verify the equation holds exactly:

$$A^2 - 3A = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = 7I \Rightarrow A^2 - 3A - 7I = O \checkmark$$

Now rearrange this identity to isolate a factor of A multiplying something equal to I – this is the step that produces A^{-1} without touching the adjoint at all:

$$A^2 - 3A = 7I \Rightarrow A(A - 3I) = 7I \Rightarrow A \left[\frac{1}{7}(A - 3I) \right] = I$$

Read off A^{-1} directly from the bracket, since this last line is exactly the definition $AA^{-1} = I$:

$$A^{-1} = \frac{1}{7}(A - 3I) = \frac{1}{7} \begin{bmatrix} 5 - 3 & 3 \\ -1 & -2 - 3 \end{bmatrix}$$

Final Answer

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

Common Mistake: Rearranging the polynomial identity incorrectly (e.g. dividing by A , which isn't valid matrix algebra) instead of factoring out A properly so that what remains multiplies to give I .

Alternate Board-Style Example

Source: Exercise 4.4, Q24(i)

Given: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$; prove $A^2 - 4A - 5I = O$, hence find A^{-1}

Solution:

Compute A^2 by direct matrix multiplication:

$$A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

Compute $4A$:

$$4A = \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix}$$

Subtract to check the identity:

$$A^2 - 4A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = 5I \Rightarrow A^2 - 4A - 5I = O \checkmark$$

Rearrange exactly as before, factoring A out on the left:

$$A^2 - 4A = 5I \Rightarrow A(A - 4I) = 5I \Rightarrow A \left[\frac{1}{5}(A - 4I) \right] = I$$

Read off A^{-1} :

$$A^{-1} = \frac{1}{5}(A - 4I) = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Final Answer

$$A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix}$$

Quick Recall: If A satisfies $A^2 - pA - qI = O$, then straight away $A^{-1} = \frac{1}{q}(A - pI)$ – no adjoint needed at all once that polynomial identity is confirmed.

Type 30

An equation of the form $BAC = I$ (or $APB = C$) is given, where some of the matrices are known and one unknown matrix must be found. The method is to isolate the unknown matrix by multiplying both sides by the appropriate inverses, on the correct side each time (matrix multiplication is not commutative, so left/right order matters).

Formula Used: If $XY = Z$ and X^{-1} exists, then $Y = X^{-1}Z$; if $YX = Z$, then $Y = ZX^{-1}$

Source: Exercise 4.4, Q28(ii)

Given: Find matrix P satisfying $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} P \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

Solution:

Name the matrices $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, so the equation reads $APB = C$ – naming them makes the multiplication order easier to track than working with raw matrices throughout.

Multiply both sides on the left by A^{-1} first, since A is the left-most matrix multiplying P :

$$A^{-1}(APB) = A^{-1}C \Rightarrow PB = A^{-1}C$$

Multiply both sides on the right by B^{-1} next, since B is now on the right of P :

$$PBB^{-1} = A^{-1}CB^{-1} \Rightarrow P = A^{-1}CB^{-1}$$

Compute A^{-1} : $|A| = 4 - 3 = 1$, so $A^{-1} = \text{adj } A = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$.

Compute B^{-1} : $|B| = 9 - 10 = -1$, $\text{adj } B = \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix}$, so $B^{-1} = \frac{1}{-1} \begin{bmatrix} -3 & -2 \\ -5 & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$.

Multiply C and B^{-1} first (working left to right avoids recomputing products):

$$CB^{-1} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} = \begin{bmatrix} 3+10 & 2+6 \\ 6-5 & 4-3 \end{bmatrix} = \begin{bmatrix} 13 & 8 \\ 1 & 1 \end{bmatrix}$$

Finally multiply A^{-1} into this result:

$$P = A^{-1}(CB^{-1}) = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 13 & 8 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 26-1 & 16-1 \\ -39+2 & -24+2 \end{bmatrix}$$

Final Answer

$$P = \begin{bmatrix} 25 & 15 \\ -37 & -22 \end{bmatrix}$$

Common Mistake: Multiplying an inverse on the wrong side (e.g. $B^{-1}A^{-1}$ instead of $A^{-1}CB^{-1}$ in the correct left-right order) – since matrices don't commute, the side and order of multiplication must exactly mirror how the unknown is sandwiched in the original equation.

Quick Recall: "Peel off" matrices from the unknown one at a time, multiplying by the inverse on whichever side that matrix sits – left-multiplying matrices on the left, right-multiplying ones on the right.

Type 31

A specific matrix is given, and the question asks to prove that its inverse equals some special related matrix – such as its own transpose ($A^{-1} = A'$) or its own square ($A^{-1} = A^2$). The method is to compute that related matrix directly and multiply it by A , checking whether the product is the identity matrix.

Formula Used: $A^{-1} = B \iff AB = I$

Source: Exercise 4.4, Q25(ii)

Given: $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$; prove $A^{-1} = A^2$

Solution:

Compute A^2 by direct matrix multiplication, since the claim to be proved is exactly that this equals A^{-1} :

$$A^2 = A \cdot A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

Rather than separately finding A^{-1} by the adjoint method, use the defining property of an inverse instead – multiply A by this computed A^2 and check if the result is I_3 , since that's a faster route to the same conclusion:

$$A \cdot A^2 = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$$

Multiply row by column carefully, one entry at a time:

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

Since $A \cdot A^2 = I_3$, this satisfies exactly the defining property of an inverse matrix.

Final Answer

Since $A \cdot A^2 = I_3$, it follows that $A^{-1} = A^2$, as required.

Common Mistake: Computing A^{-1} the long way via the adjoint formula and then separately computing A^2 to compare them, instead of the shorter route of directly checking $A \cdot A^2 = I$ – both work, but the direct check is faster and less error-prone for a 3×3 matrix.

Quick Recall: To prove A^{-1} equals some matrix B , don't compute A^{-1} from scratch – just multiply A by B and check the product is I .

V. Solving Systems of Linear Equations by Matrix Method

Type 32

A system of two linear equations in two unknowns is written as $AX = B$, and solved for X by computing A^{-1} and multiplying it into B – this is the direct matrix-method analogue of solving simultaneous equations.

Formula Used: $AX = B \Rightarrow X = A^{-1}B, \quad |A| \neq 0$

Source: Exercise 4.5, Q1(i)

Given: $5x + 2y = 4, \quad 7x + 3y = 5$

Solution:

Write the system in matrix form $AX = B$, reading the coefficients straight off the equations – this is the setup step that turns "simultaneous equations" into a matrix problem:

$$A = \begin{bmatrix} 5 & 2 \\ 7 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

Compute $|A|$ to confirm a unique solution exists before going any further:

$$|A| = (5)(3) - (2)(7) = 15 - 14 = 1 \neq 0$$

Find A^{-1} using the 2×2 shortcut, since $|A| = 1$ this is especially simple:

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix}$$

Multiply A^{-1} into B to get X , since $X = A^{-1}B$ is exactly what solves the system:

$$X = \begin{bmatrix} 3 & -2 \\ -7 & 5 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 - 10 \\ -28 + 25 \end{bmatrix}$$

Final Answer

$$x = 2, \quad y = -3$$

Common Mistake: Setting up B with the wrong sign or order (e.g. writing the equations as $Ax = B$ with the constants on the wrong side) – always move every equation into the exact form $ax + by = c$ before reading off A and B .

Quick Recall: Read coefficients into A , constants into B , and solve with $X = A^{-1}B$ – exactly like $x = b/a$ for a single equation, but with matrices.

Type 33

The question asks not to solve a system, but to determine whether it has a solution at all – i.e. whether it is consistent or inconsistent. This is checked using $|A|$, and (if $|A| = 0$) further checked using whether $(adj A)B$ is the zero matrix.

Formula Used: $|A| \neq 0 \Rightarrow$ consistent (unique solution)

Formula Used: $|A| = 0$ and $(adj A)B = O \Rightarrow$ consistent (infinite solutions); $|A| = 0$ and $(adj A)B \neq O \Rightarrow$ inconsistent

Source: Exercise 4.5, Q2(i) and Q2(ii)

Given: (i) $x + 2y = 2, 2x + 3y = 3$ (ii) $x + 2y = 9, 2x + 4y = 7$

Solution:

Case: $|A| \neq 0$ – Q2(i)

Write the coefficient matrix and compute its determinant, since that's the first and quickest test for consistency:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}, \quad |A| = 3 - 4 = -1 \neq 0$$

Since $|A| \neq 0$, A^{-1} exists, so $X = A^{-1}B$ gives a unique solution – no further check is needed.

Final Answer

The system is consistent (unique solution).

Case: $|A| = 0$ – Q2(ii)

Compute the determinant for this second system:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad |A| = 4 - 4 = 0$$

Since $|A| = 0$, a unique solution is ruled out, but the system could still be consistent with infinitely many solutions – so find $adj A$ and check $(adj A)B$ against the zero matrix, since that's what distinguishes the two remaining possibilities:

$$adj A = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$$

$$(adj A)B = \begin{bmatrix} 4(9) - 2(7) \\ -2(9) + 1(7) \end{bmatrix} = \begin{bmatrix} 22 \\ -11 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since $(adj A)B \neq O$, the system has no solution at all.

Final Answer

The system is inconsistent.

Common Mistake: Stopping as soon as $|A| = 0$ and declaring the system "inconsistent" without checking $(adj A)B$ – a zero determinant alone does not distinguish between "no solution" and "infinitely many solutions".

Quick Recall: $|A| \neq 0$ means one clean answer; if $|A| = 0$, check $(adj A)B$ – zero means infinitely many solutions, nonzero means no solution at all.

Type 34

A system of three linear equations in three unknowns is solved by finding A^{-1} (via the adjoint) and computing $X = A^{-1}B$ – the natural extension of Type 32 to three variables, and the most common long-answer format built on everything learned about adjoints and inverses.

★ **Board Favorite** – Solving a 3×3 system by the matrix method is one of the most reliably tested long-answer questions across ISC papers on this chapter – a standard high-yield pattern (exact past years not confirmed from this book, though several sourced questions here are explicitly marked ISC in the textbook).

Formula Used: $AX = B \Rightarrow X = A^{-1}B = \frac{1}{|A|}(adj A)B$

Source: Exercise 4.5, Q9

Given: $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$

Solution:

Write the system as $AX = B$:

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}, \quad B = \begin{bmatrix} -4 \\ 2 \\ 11 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1, to confirm a unique solution exists:

$$|A| = 1(-12 + 6) - 2(-8 - 6) + (-3)(-6 - 9) = -6 + 28 + 45 = 67$$

Find every cofactor systematically, since $adj A$ needs all nine of them before it can be assembled:

$$A_{11} = -6, A_{12} = 14, A_{13} = -15, A_{21} = 17, A_{22} = 5, A_{23} = 9, A_{31} = 13, A_{32} = -8, A_{33} = -1$$

Transpose the cofactor matrix to build $adj A$:

$$adj A = \begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$$

Multiply $(adj A)B$ first, then divide by $|A|$ at the very end – this order avoids fractions until the last possible step:

$$(adj A)B = \begin{bmatrix} -6(-4) + 17(2) + 13(11) \\ 14(-4) + 5(2) - 8(11) \\ -15(-4) + 9(2) - 1(11) \end{bmatrix} = \begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix}$$

Divide every entry by $|A| = 67$ to get X :

$$X = \frac{1}{67} \begin{bmatrix} 201 \\ -134 \\ 67 \end{bmatrix}$$

Final Answer

$$x = 3, y = -2, z = 1$$

Common Mistake: Making a sign or arithmetic slip in even one of the nine cofactors – since every cofactor feeds into $adj A$, a single error there silently corrupts the whole final answer, so it's worth sanity-checking $A(adj A) = |A|I$ before proceeding.

Alternate Board-Style Example

Source: Exercise 4.5, Q4(ii)

$$\text{Given: } x + y + z = 6, \quad x - y + z = 2, \quad 2x + y - z = 1$$

Solution:

Set up A and B from the coefficients:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 1(1 - 1) - 1(-1 - 2) + 1(1 + 2) = 0 + 3 + 3 = 6$$

Find the cofactors and assemble $adj A$ by transposing:

$$adj A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

Multiply $(adj A)B$:

$$(adj A)B = \begin{bmatrix} 0(6) + 2(2) + 2(1) \\ 3(6) - 3(2) + 0(1) \\ 3(6) + 1(2) - 2(1) \end{bmatrix} = \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix}$$

Divide by $|A| = 6$:

$$X = \frac{1}{6} \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix}$$

Final Answer

$$x = 1, y = 2, z = 3$$

Quick Recall: Same three steps as the 2×2 case, just bigger: find $|A|$, build $\text{adj } A$ from all nine cofactors, then $X = \frac{1}{|A|}(\text{adj } A)B$.

Type 35

A system is given in terms of $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ rather than x, y, z directly. The trick is to substitute $u = \frac{1}{x}$, $v = \frac{1}{y}$, $w = \frac{1}{z}$ to turn it into an ordinary linear system, solve by the matrix method as usual, and then invert the substitution at the very end.

Formula Used: $u = \frac{1}{x}$, $v = \frac{1}{y}$, $w = \frac{1}{z} \Rightarrow$ ordinary linear system in u, v, w

Source: Exercise 4.5, Q5

$$\text{Given: } \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 9, \quad \frac{2}{x} + \frac{5}{y} + \frac{7}{z} = 52, \quad \frac{2}{x} + \frac{1}{y} - \frac{1}{z} = 0$$

Solution:

Substitute $u = \frac{1}{x}$, $v = \frac{1}{y}$, $w = \frac{1}{z}$, since this converts the reciprocal system into an ordinary linear one that the matrix method can handle directly:

$$u + v + w = 9, \quad 2u + 5v + 7w = 52, \quad 2u + v - w = 0$$

Set up A and B from these coefficients:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & 7 \\ 2 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 9 \\ 52 \\ 0 \end{bmatrix}$$

Compute $|A|$:

$$|A| = 1(-5 - 7) - 1(-2 - 14) + 1(2 - 10) = -12 + 16 - 8 = -4$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} -12 & 2 & 2 \\ 16 & -3 & -5 \\ -8 & 1 & 3 \end{bmatrix}$$

Multiply $(adj A)B$ and divide by $|A| = -4$:

$$(adj A)B = \begin{bmatrix} -12(9) + 2(52) + 2(0) \\ 16(9) - 3(52) - 5(0) \\ -8(9) + 1(52) + 3(0) \end{bmatrix} = \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \frac{1}{-4} \begin{bmatrix} -4 \\ -12 \\ -20 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

Finally, invert the substitution to recover x, y, z – this last step is the one most easily forgotten, since the matrix method only ever solves for u, v, w :

$$x = \frac{1}{u} = 1, \quad y = \frac{1}{v} = \frac{1}{3}, \quad z = \frac{1}{w} = \frac{1}{5}$$

Final Answer

$$x = 1, \quad y = \frac{1}{3}, \quad z = \frac{1}{5}$$

Common Mistake: Solving for u, v, w correctly and then forgetting to invert back to x, y, z at the end – the final answer must be in terms of the original variables, not the substituted ones.

Quick Recall: Whenever a system has $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ instead of x, y, z , substitute u, v, w for the reciprocals, solve normally, then flip back at the very last step.

Type 36

A real-world scenario (a shop, a cafe, a mixture of quantities) is described in words, and must first be translated into a system of linear equations before being solved by the matrix method. The translation step – carefully reading each sentence into one equation – is as important as the solving itself.

★ **Board Favorite** – Word problems are a guaranteed long-answer question on this chapter – the matrix method is one of the standard ways ISC tests "translate words into equations, then solve" across board papers – a standard high-yield pattern (exact past years not confirmed from this book).

Formula Used: $AX = B \Rightarrow X = A^{-1}B$

Source: Exercise 4.5, Q8

Given: 100 items sold in total (burgers, samosas, juice bottles); burgers = $3 \times$ samosas; samosas = juice + 10

Solution:

Let x, y, z denote the number of burgers, samosas, and juice bottles sold, so each sentence in the problem can be translated one at a time into an equation:

"100 items in total" becomes:

$$x + y + z = 100$$

"burgers were thrice the samosas" becomes:

$$x = 3y \Rightarrow x - 3y = 0$$

"samosas were 10 more than juice bottles" becomes:

$$y = z + 10 \Rightarrow y - z = 10$$

Set up A and B from these three equations:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -3 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 100 \\ 0 \\ 10 \end{bmatrix}$$

Compute $|A|$:

$$|A| = 1(3 - 0) - 1(-1 - 0) + 1(1 - 0) = 3 + 1 + 1 = 5$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 3 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & -1 & -4 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = 5$:

$$(\text{adj } A)B = \begin{bmatrix} 3(100) + 1(0) + 3(10) \\ 1(100) - 1(0) + 1(10) \\ 1(100) - 1(0) - 4(10) \end{bmatrix} = \begin{bmatrix} 330 \\ 110 \\ 60 \end{bmatrix}$$

$$X = \frac{1}{5} \begin{bmatrix} 330 \\ 110 \\ 60 \end{bmatrix}$$

Final Answer

Burgers = 66, Samosas = 22, Juice bottles = 12

Common Mistake: Setting up an equation in the wrong direction (e.g. writing $3x = y$ instead of $x = 3y$ for "burgers were thrice the samosas") – always re-read the sentence and check which quantity is the multiple of which.

Alternate Board-Style Example

Source: Exercise 4.5, Q7

Given: The sum of three numbers is 6. Thrice the third number added to the second gives 11. Adding the first and third numbers gives double the second number.

Solution:

Let the three numbers be x, y, z , and translate each sentence in turn.

"The sum of three numbers is 6" becomes:

$$x + y + z = 6$$

"Thrice the third number, added to the second, gives 11" – read carefully, this means $3z + y = 11$, not $3(y + z) = 11$, since only the third number is being tripled:

$$y + 3z = 11$$

"Adding the first and third numbers gives double the second number" becomes:

$$x + z = 2y \Rightarrow x - 2y + z = 0$$

Set up A and B :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 1 & -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 6 \\ 11 \\ 0 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 1(1 + 6) - 1(0 - 3) + 1(0 - 1) = 7 + 3 - 1 = 9$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 7 & -3 & 2 \\ 3 & 0 & -3 \\ -1 & 3 & 1 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = 9$:

$$(\text{adj } A)B = \begin{bmatrix} 7(6) - 3(11) + 2(0) \\ 3(6) + 0(11) - 3(0) \\ -1(6) + 3(11) + 1(0) \end{bmatrix} = \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix}$$

$$X = \frac{1}{9} \begin{bmatrix} 9 \\ 18 \\ 27 \end{bmatrix}$$

Final Answer

The three numbers are 1, 2, 3.

Common Mistake: Misreading "thrice the third number added to the second" as $3(y + z) = 11$ instead of $(3z) + y = 11$ – in word problems, exactly which quantity gets multiplied is easy to misplace, so re-read the sentence slowly before writing the equation.

Quick Recall: Translate each sentence into one equation, being careful about which quantity is "thrice" or "more than" which – then solve exactly like any other 3×3 system.

Extra Practice – Word Problems

The exercise itself contains only the two word problems above. The five that follow are additional practice problems, written in the same exam style and solved with the same matrix method – they are *not* from ML Aggarwal, and are included only because more word-problem practice was requested. Each is solved and double-checked exactly like a textbook question.

Source: Original practice problem 1 – not from ML Aggarwal

Given: The sum of the present ages of a father, mother and son is 105 years. Ten years ago, the father's age was thrice the son's age. Five years from now, the mother's age will be twice the son's age. Find their present ages.

Solution:

Let F, M, S denote the present ages of the father, mother and son, and translate each sentence in turn.

"The sum of their present ages is 105" becomes:

$$F + M + S = 105$$

"Ten years ago, the father's age was thrice the son's age" – both ages must be reduced by 10 before comparing, since "ten years ago" shifts both quantities back in time together:

$$F - 10 = 3(S - 10) \Rightarrow F - 3S = -20$$

"Five years from now, the mother's age will be twice the son's age", similarly with both ages shifted forward by 5:

$$M + 5 = 2(S + 5) \Rightarrow M - 2S = 5$$

Set up A and B from these three equations:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & -3 \\ 0 & 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 105 \\ -20 \\ 5 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 1(0 + 3) - 1(-2 - 0) + 1(1 - 0) = 3 + 2 + 1 = 6$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 3 & 3 & -3 \\ 2 & -2 & 4 \\ 1 & -1 & -1 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = 6$:

$$(\text{adj } A)B = \begin{bmatrix} 3(105) + 3(-20) - 3(5) \\ 2(105) - 2(-20) + 4(5) \\ 1(105) - 1(-20) - 1(5) \end{bmatrix} = \begin{bmatrix} 240 \\ 270 \\ 120 \end{bmatrix}$$

$$X = \frac{1}{6} \begin{bmatrix} 240 \\ 270 \\ 120 \end{bmatrix}$$

Final Answer

Father = 40 years, Mother = 45 years, Son = 20 years.

Common Mistake: Writing "ten years ago the father was thrice the son's age" as $F = 3S - 10$ instead of $F - 10 = 3(S - 10)$ – the "ago"/"hence" shift applies to *both* ages before the multiplication, not just one side.

Source: Original practice problem 2 – not from ML Aggarwal

Given: 3 chairs, 2 tables and 1 lamp together cost Rs.1350. 1 chair, 3 tables and 2 lamps together cost Rs.1400. 2 chairs, 1 table and 3 lamps together cost Rs.1150. Find the cost of each item.

Solution:

Let x, y, z be the cost (in Rs.) of one chair, one table and one lamp respectively, and translate each sentence into an equation:

$$3x + 2y + z = 1350, \quad x + 3y + 2z = 1400, \quad 2x + y + 3z = 1150$$

Set up A and B :

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 3 & 2 \\ 2 & 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1350 \\ 1400 \\ 1150 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 3(9 - 2) - 2(3 - 4) + 1(1 - 6) = 21 + 2 - 5 = 18$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 7 & -5 & 1 \\ 1 & 7 & -5 \\ -5 & 1 & 7 \end{bmatrix}$$

Multiply $(adj A)B$ and divide by $|A| = 18$:

$$(adj A)B = \begin{bmatrix} 7(1350) - 5(1400) + 1(1150) \\ 1(1350) + 7(1400) - 5(1150) \\ -5(1350) + 1(1400) + 7(1150) \end{bmatrix} = \begin{bmatrix} 3600 \\ 5400 \\ 2700 \end{bmatrix}$$

$$X = \frac{1}{18} \begin{bmatrix} 3600 \\ 5400 \\ 2700 \end{bmatrix}$$

Final Answer

Chair = Rs. 200, Table = Rs. 300, Lamp = Rs. 150.

Common Mistake: Mixing up which coefficient goes with which unknown when three similar items are all mentioned in every sentence – write x, y, z for chair, table, lamp explicitly first, and keep that order fixed in every equation.

Source: Original practice problem 3 – not from ML Aggarwal

Given: A person invests a total of Rs. 30,000 in three schemes A, B and C, paying simple interest at 4%, 5% and 6% per annum respectively. The total interest earned in one year is Rs. 1520. Also, the amount invested in schemes A and B together equals twice the amount invested in scheme C. Find the amount invested in each scheme.

Solution:

Let x, y, z be the amounts (in Rs.) invested in schemes A, B and C.

”Total invested is 30,000” becomes:

$$x + y + z = 30000$$

”Total interest earned is 1520” – multiply the percentages through by 100 on both sides to clear decimals before setting up the matrix:

$$0.04x + 0.05y + 0.06z = 1520 \Rightarrow 4x + 5y + 6z = 152000$$

”Amount in A and B together equals twice the amount in C” becomes:

$$x + y = 2z \Rightarrow x + y - 2z = 0$$

Set up A and B :

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 30000 \\ 0 \\ 152000 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 1(6 + 10) - 1(6 + 8) + 1(5 - 4) = 16 - 14 + 1 = 3$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 16 & -1 & -3 \\ -14 & 2 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = 3$:

$$(\text{adj } A)B = \begin{bmatrix} 16(30000) - 1(0) - 3(152000) \\ -14(30000) + 2(0) + 3(152000) \\ 1(30000) - 1(0) + 0(152000) \end{bmatrix} = \begin{bmatrix} 24000 \\ 36000 \\ 30000 \end{bmatrix}$$

$$X = \frac{1}{3} \begin{bmatrix} 24000 \\ 36000 \\ 30000 \end{bmatrix}$$

Final Answer

Scheme A = Rs. 8000, Scheme B = Rs. 12000, Scheme C = Rs. 10000.

Common Mistake: Forgetting to clear the percentage decimals (leaving 0.04, 0.05, 0.06 as the matrix entries) – this doesn't break the method, but it drags decimals through every cofactor unnecessarily; multiplying the interest equation by 100 first keeps all working in whole numbers.

Source: Original practice problem 4 – not from ML Aggarwal

Given: A three-digit number has digits whose sum is 12. The middle (tens) digit equals the sum of the other two digits. If the digits are reversed, the new number is 594 less than the original number. Find the number.

Solution:

Let the hundreds, tens and units digits be x, y, z , so the number itself is $100x + 10y + z$.

"Sum of the digits is 12" becomes:

$$x + y + z = 12$$

"The middle digit equals the sum of the other two" becomes:

$$y = x + z \Rightarrow -x + y - z = 0$$

"Reversing the digits reduces the number by 594" – the reversed number is $100z + 10y + x$, so subtract it from the original and set the difference to 594:

$$(100x + 10y + z) - (100z + 10y + x) = 594 \Rightarrow 99x - 99z = 594 \Rightarrow x - z = 6$$

Set up A and B , writing the third equation with a 0 coefficient for y since it doesn't appear in it:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & -1 \\ 1 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 12 \\ 0 \\ 6 \end{bmatrix}$$

Compute $|A|$ by expanding along Row 1:

$$|A| = 1(-1 - 0) - 1(1 + 1) + 1(0 - 1) = -1 - 2 - 1 = -4$$

Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} -1 & 1 & -2 \\ -2 & -2 & 0 \\ -1 & 1 & 2 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = -4$:

$$(\text{adj } A)B = \begin{bmatrix} -1(12) + 1(0) - 2(6) \\ -2(12) - 2(0) + 0(6) \\ -1(12) + 1(0) + 2(6) \end{bmatrix} = \begin{bmatrix} -24 \\ -24 \\ 0 \end{bmatrix}$$

$$X = \frac{1}{-4} \begin{bmatrix} -24 \\ -24 \\ 0 \end{bmatrix}$$

Final Answer

Digits are $x = 6$, $y = 6$, $z = 0$, so the number is **660**.

Common Mistake: Writing the reversed number as $z + 10y + 100x$ (i.e. accidentally keeping the original place values) instead of correctly swapping hundreds and units to get $100z + 10y + x$ – the tens digit stays in the middle either way, only the hundreds and units digits swap.

Source: Original practice problem 5 – not from ML Aggarwal

Given: Two stations A and B are 300 km apart. Two trains start at the same time, one from each station, travelling towards each other, and meet after 3 hours. If instead both trains travel in the same direction (both starting from A and B, the one from A being faster), the faster train catches up to the slower one after 15 hours. Find the speed of each train.

Solution:

Let a = speed of the train starting from A, and b = speed of the train starting from B (in km/h), and translate each scenario using relative speed – this is a 2×2 system, showing the same matrix method works regardless of how many unknowns there are.

”Travelling towards each other, they meet after 3 hours” – when moving towards each other, their speeds add, and together they cover the full 300 km gap:

$$(a + b) \times 3 = 300 \Rightarrow a + b = 100$$

”Travelling in the same direction, the faster catches up after 15 hours” – when moving the same way, only the *difference* in speed closes the gap, and it must close the same 300 km:

$$(a - b) \times 15 = 300 \Rightarrow a - b = 20$$

Set up A and B :

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$$

Compute $|A|$:

$$|A| = (1)(-1) - (1)(1) = -2$$

Find A^{-1} using the 2×2 shortcut:

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

Multiply $A^{-1}B$:

$$X = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 100 \\ 20 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} -120 \\ -80 \end{bmatrix}$$

Final Answer

Train from A = 60 km/h, Train from B = 40 km/h.

Common Mistake: Using the speed *sum* for the same-direction (catch-up) case instead of the speed *difference* – "towards each other" always adds speeds, "same direction, catching up" always subtracts them.

Quick Recall: Speed/distance word problems translate into linear equations through relative speed: speeds *add* when travelling towards each other, and *subtract* when travelling the same way – then solve exactly like any other matrix-method system.

Type 37

A homogeneous system $AX = O$ is given (every equation equals zero on the right). Such a system is always consistent, but whether it has *only* the trivial solution $x = y = z = 0$ or infinitely many solutions depends entirely on whether $|A|$ is nonzero or zero.

Formula Used: $AX = O : |A| \neq 0 \Rightarrow \text{only } X = O; |A| = 0 \Rightarrow \text{infinitely many solutions}$

Source: Exercise 4.5, Q15(ii) and Q16(i)

Given: (a) $3x + y - 2z = 0, x + y + z = 0, x - 2y + z = 0$ (b) $2x - 3y - z = 0, x + 3y - 2z = 0, x - 3y = 0$

Solution:

Case: $|A| \neq 0$ – only the trivial solution (Ex 4.5, Q15(ii))

Compute $|A|$ for the coefficient matrix of this homogeneous system:

$$A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 1 & 1 \\ 1 & -2 & 1 \end{bmatrix}, \quad |A| = 3(1 + 2) - 1(1 - 1) - 2(-2 - 1) = 9 - 0 + 6 = 15$$

Since $|A| = 15 \neq 0$, A^{-1} exists, so the only solution to $AX = O$ is $X = A^{-1}O = O$ – a nonzero determinant always forces the trivial solution for a homogeneous system.

Final Answer

$x = 0, y = 0, z = 0$ (only the trivial solution).

Case: $|A| = 0$ – infinitely many solutions (Ex 4.5, Q16(i))

Compute $|A|$ for this second homogeneous system:

$$A = \begin{bmatrix} 2 & -3 & -1 \\ 1 & 3 & -2 \\ 1 & -3 & 0 \end{bmatrix}, \quad |A| = 2(0 - 6) + 3(0 + 2) - 1(-3 - 3) = -12 + 6 + 6 = 0$$

Since $|A| = 0$, infinitely many solutions exist, so express two of the variables in terms of the third using the original equations directly (matrix inversion isn't possible here, since A^{-1} doesn't exist): from $x - 3y = 0$, $x = 3y$; substituting into $x + 3y - 2z = 0$ gives $6y = 2z$, so $z = 3y$.

Let $y = \frac{k}{3}$ (chosen so that x and z come out as clean multiples of k):

$$x = 3y = k, \quad z = 3y = k$$

Final Answer

$x = k, y = \frac{k}{3}, z = k$, where k is any real number.

Common Mistake: Trying to compute A^{-1} when $|A| = 0$ – once the determinant is zero, the matrix method itself breaks down, and the system must instead be solved by directly relating the variables through the original equations.

Quick Recall: For $AX = O$: check $|A|$ first – nonzero means $x = y = z = 0$ is the only answer; zero means infinitely many solutions exist, found by expressing variables in terms of a parameter.

Type 38

A system contains unknown coefficients in symbolic form, and the question first asks for the condition under which the system has a unique solution, before actually solving it for the given numbers. The condition is always $|A| \neq 0$; the solving itself uses the ordinary matrix method.

Formula Used: Unique solution to $AX = B \iff |A| \neq 0$

Source: Exercise 4.5, Q6

Given: $x + 2y - 2z + 5 = 0$, $-x + 3y + 4 = 0$, $-2y + z - 4 = 0$

Solution:

Rearrange every equation into the standard $AX = B$ form first, moving constants to the right-hand side – this is necessary since the equations are currently written with everything on one side:

$$x + 2y - 2z = -5, \quad -x + 3y + 0z = -4, \quad 0x - 2y + z = 4$$

$$A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -5 \\ -4 \\ 4 \end{bmatrix}$$

State the condition for a unique solution before solving anything – this is exactly what the question asks for first:

$$\text{Unique solution exists} \iff |A| \neq 0$$

Compute $|A|$ to confirm the condition actually holds for this system:

$$|A| = 1(3 - 0) - 2(-1 - 0) - 2(2 - 0) = 3 + 2 - 4 = 1 \neq 0$$

Since the condition is satisfied, proceed to solve using A^{-1} . Find the cofactors and build $\text{adj } A$:

$$\text{adj } A = \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

Multiply $(\text{adj } A)B$ and divide by $|A| = 1$:

$$(\text{adj } A)B = \begin{bmatrix} 3(-5) + 2(-4) + 6(4) \\ 1(-5) + 1(-4) + 2(4) \\ 2(-5) + 2(-4) + 5(4) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

Final Answer

Unique solution exists since $|A| = 1 \neq 0$; $x = 1$, $y = -1$, $z = 2$.

Common Mistake: Jumping straight to solving without first explicitly stating the condition $|A| \neq 0$ – when a question specifically asks for the condition, stating it is a method mark on its own, separate from the final numeric solution.

Quick Recall: "Condition for unique solution" always means $|A| \neq 0$ – state that first, verify it holds, then solve exactly as in any other matrix-method system.

VI. Higher-Order / Competency-Focused Questions

Type 39

This is a genuinely tougher variant of the "prove an identity" pattern: the determinant's entries are built from $a^2, b^2, c^2, ab, bc, ca$ rather than plain letters, and the book itself flags it as a Competency Focused Question. Two different determinants must both be shown to reduce to the same expression – one by careful full expansion (watching a higher-degree cancellation happen), the other by a row-operation shortcut.

★ **Board Favorite** – Competency Focused Questions built from a^2, b^2, c^2 -type entries are exactly the style ISC has been leaning into in recent papers to test genuine understanding rather than rote steps – a standard high-yield pattern for the toughest 1-2 questions on this chapter (exact past years not confirmed from this book, though the book itself tags this one as Competency Focused).

Formula Used: $\Delta = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$

Source: Exercise 4.2, Q30 (Competency Focused Question)

$$\text{Given: } \begin{vmatrix} a^2 + 1 & ab & ac \\ ab & b^2 + 1 & bc \\ ac & bc & c^2 + 1 \end{vmatrix} = \begin{vmatrix} a^2 + 1 & b^2 & c^2 \\ a^2 & b^2 + 1 & c^2 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix} = 1 + a^2 + b^2 + c^2$$

Solution:

Start with the left-hand determinant and expand fully along Row 1, since no obvious row/column shortcut is visible here and the higher-degree terms genuinely need to be tracked carefully:

$$\Delta_1 = (a^2 + 1)[(b^2 + 1)(c^2 + 1) - b^2c^2] - ab[ab(c^2 + 1) - bc \cdot ac] + ac[ab \cdot bc - (b^2 + 1)ac]$$

Simplify the first bracket, since $(b^2 + 1)(c^2 + 1) - b^2c^2$ collapses nicely once expanded:

$$(b^2 + 1)(c^2 + 1) - b^2c^2 = b^2c^2 + b^2 + c^2 + 1 - b^2c^2 = b^2 + c^2 + 1$$

So the first term becomes $(a^2 + 1)(b^2 + c^2 + 1) = a^2b^2 + a^2c^2 + a^2 + b^2 + c^2 + 1$.

Simplify the second bracket similarly:

$$ab(c^2 + 1) - abc^2 = abc^2 + ab - abc^2 = ab \Rightarrow \text{second term} = -ab(ab) = -a^2b^2$$

Simplify the third bracket:

$$ab^2c - (b^2 + 1)ac = ab^2c - ab^2c - ac = -ac \Rightarrow \text{third term} = ac(-ac) = -a^2c^2$$

Add all three terms together – this is the step that makes the question "hard": the a^2b^2 and a^2c^2 terms appear with opposite signs in different terms and must be spotted and cancelled correctly:

$$\Delta_1 = [a^2b^2 + a^2c^2 + a^2 + b^2 + c^2 + 1] - a^2b^2 - a^2c^2 = a^2 + b^2 + c^2 + 1$$

Now handle the right-hand (middle) determinant separately, this time using row operations instead of full expansion. Subtract Row 3 from Row 1 and from Row 2, since that's what strips away the repeated a^2, b^2, c^2 pattern along each row:

$$R_1 \rightarrow R_1 - R_3 : (1, 0, -1), \quad R_2 \rightarrow R_2 - R_3 : (0, 1, -1)$$

The matrix becomes:

$$\begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ a^2 & b^2 & c^2 + 1 \end{vmatrix}$$

Expand along Row 1, where two of the three entries are now conveniently 0:

$$\Delta_2 = 1[(c^2 + 1) - (-b^2)] - 0 + (-1)[0 - a^2] = (c^2 + 1 + b^2) + a^2$$

Final Answer

Both determinants equal $1 + a^2 + b^2 + c^2$, and are therefore equal to each other, as required.

Common Mistake: In the full-expansion route, missing that a^2b^2 and a^2c^2 appear in *two* places with opposite sign (once inside the first bracket's expansion, once as a standalone term) – losing track of either occurrence gives an answer with leftover a^2b^2 or a^2c^2 terms that shouldn't be there.

Quick Recall: When a determinant looks like "identity matrix plus an outer-product pattern" (a^2, b^2, c^2 on the diagonal, cross-products like ab, bc, ca off it), the value is almost always 1 plus the sum of squares – but you still have to earn that answer through careful expansion or a row-difference trick.

Type 40

This type moves the determinant idea into coordinate geometry at a noticeably harder level: instead of computing a numeric area, an algebraic identity connecting the *square* of an area-style determinant to the side of an equilateral triangle must be proved. It combines the area formula, a column-scaling property, and squaring both sides.

Formula Used: Area of equilateral triangle, side $a = \frac{\sqrt{3}}{4}a^2$

Formula Used: Area = $\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

Source: Example 7, Section 4.3 (Competency Focused Question)

Given: $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are vertices of an equilateral triangle with each side = a ; prove $\begin{vmatrix} x_1 & y_1 & 2 \\ x_2 & y_2 & 2 \\ x_3 & y_3 & 2 \end{vmatrix}^2 = 3a^4$

Solution:

Start from the standard area formula applied to this triangle, since that is the only piece of given information connecting the coordinates to the side length a :

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Substitute the known area of an equilateral triangle with side a , since that's the geometric fact this problem hinges on:

$$\frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{\sqrt{3}}{4} a^2$$

Multiply both sides by 2 to clear the leading fraction on the left:

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{\sqrt{3}}{2} a^2$$

Multiply both sides by 2 again, but this time interpret it as scaling Column 3 (the column of 1's) by 2 – since scaling one column of a determinant by 2 scales the whole determinant by 2, this is exactly what turns the "1's" column into the "2's" column asked for in the question:

$$\begin{vmatrix} x_1 & y_1 & 2 \\ x_2 & y_2 & 2 \\ x_3 & y_3 & 2 \end{vmatrix} = 2 \times \frac{\sqrt{3}}{2} a^2 = \sqrt{3} a^2$$

Square both sides, since the required identity is stated in terms of the determinant squared – this last step is what most students forget to do even after getting the line above correct:

$$\begin{vmatrix} x_1 & y_1 & 2 \\ x_2 & y_2 & 2 \\ x_3 & y_3 & 2 \end{vmatrix}^2 = (\sqrt{3} a^2)^2 = 3a^4$$

Final Answer

Hence the identity is proved: $\begin{vmatrix} x_1 & y_1 & 2 \\ x_2 & y_2 & 2 \\ x_3 & y_3 & 2 \end{vmatrix}^2 = 3a^4$.

Common Mistake: Treating the "column of 2's" as if it needed a completely new determinant setup, instead of recognising it as the ordinary area determinant with Column 3 simply scaled by 2 – this is a column-scaling property question in geometric disguise.

Quick Recall: Whenever a determinant has a column of 2's (or any constant other than 1) in the area formula's usual "1's" position, that's just the area determinant with one column scaled – convert back to the familiar area formula first, then scale and square at the end.

Type 41

A specific 3×3 matrix (with fractional, symmetric-looking entries) must be shown to satisfy $A^{-1} = A'$ (its own transpose) – a harder variant of Type 31, since computing A^{-1} the long way via the adjoint would involve messy fractions throughout. The efficient route recognises that A is built from a matrix whose rows are mutually perpendicular (an orthogonal-matrix pattern), so AA' collapses to a multiple of I almost immediately.

Formula Used: $A^{-1} = B \iff AB = I$

Source: Exercise 4.4, Q25(i)

Given: $A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$; prove $A^{-1} = A'$

Solution:

Write $A = \frac{1}{9}M$ where $M = \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, so that $A' = \frac{1}{9}M^T$ – pulling the $\frac{1}{9}$ out front keeps the arithmetic in whole numbers until the very last step.

Rather than computing A^{-1} from the adjoint (which would drag fractions through nine separate cofactors), use the defining property instead: compute AA' directly and check whether it equals I , since $AA' = I$ is exactly the statement $A^{-1} = A'$ requires:

$$AA' = \frac{1}{81}MM^T$$

Compute the entries of MM^T by taking dot products of the rows of M with each other, since $(MM^T)_{ij}$ is exactly Row i of M dotted with Row j of M – this is faster than a full matrix multiplication because most of the needed structure is just "is this row perpendicular to that row":

$$(MM^T)_{11} = (-8)^2 + 1^2 + 4^2 = 64 + 1 + 16 = 81$$

$$(MM^T)_{22} = 4^2 + 4^2 + 7^2 = 16 + 16 + 49 = 81$$

$$(MM^T)_{33} = 1^2 + (-8)^2 + 4^2 = 1 + 64 + 16 = 81$$

Check the off-diagonal (cross) dot products, since these need to vanish for MM^T to be a clean multiple of I :

$$(MM^T)_{12} = (-8)(4) + (1)(4) + (4)(7) = -32 + 4 + 28 = 0$$

$$(MM^T)_{13} = (-8)(1) + (1)(-8) + (4)(4) = -8 - 8 + 16 = 0$$

$$(MM^T)_{23} = (4)(1) + (4)(-8) + (7)(4) = 4 - 32 + 28 = 0$$

Every off-diagonal entry vanishes and every diagonal entry is 81, so:

$$MM^T = 81 I \Rightarrow AA' = \frac{1}{81}(81 I) = I$$

Final Answer

Since $AA' = I$, it follows that $A^{-1} = A'$, as required.

Common Mistake: Attempting the full adjoint-formula route on a matrix with a $\frac{1}{9}$ factor and nine fractional cofactors – this is technically possible but extremely error-prone; recognising the "rows are mutually perpendicular, same length" pattern and checking $AA' = I$ directly is both faster and far less likely to go wrong.

Quick Recall: To prove $A^{-1} = A'$ for a matrix like this, don't invert – multiply A by its own transpose and check you get I ; if every row has the same length and every pair of rows is perpendicular (dot product 0), that check passes instantly.